

Chapter 28 Condensed Matter

28.1 Spatial orders in condensed matter

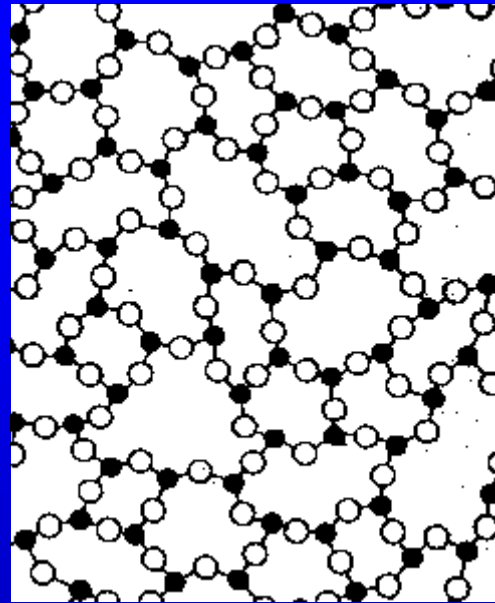
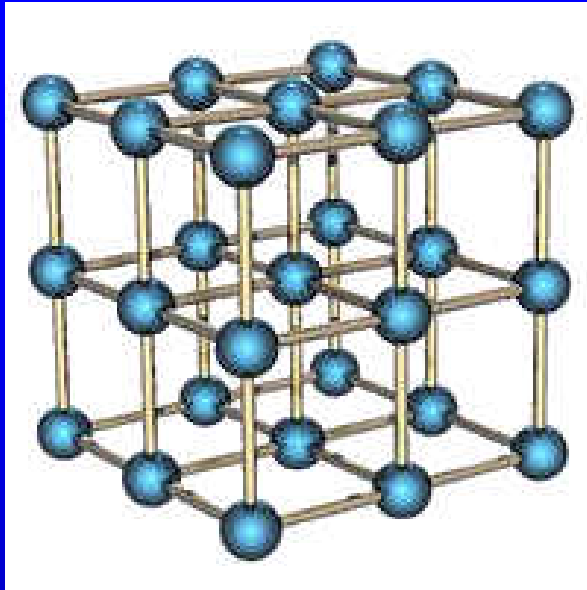
gas ~ **disordered**

liquid ~ fixed coordination number --(number of nearest neighbors) , average distance to the nearest neighbor has a characteristic value : **short-range order**

crystal~equilibrium positions of all atoms (ions or molecules) are fixed on a **lattice**

translational symmetry **periodicity**

long-range position order **bond orientation order**



The amorphous solid (glass) does not have long-range order but short-range order.

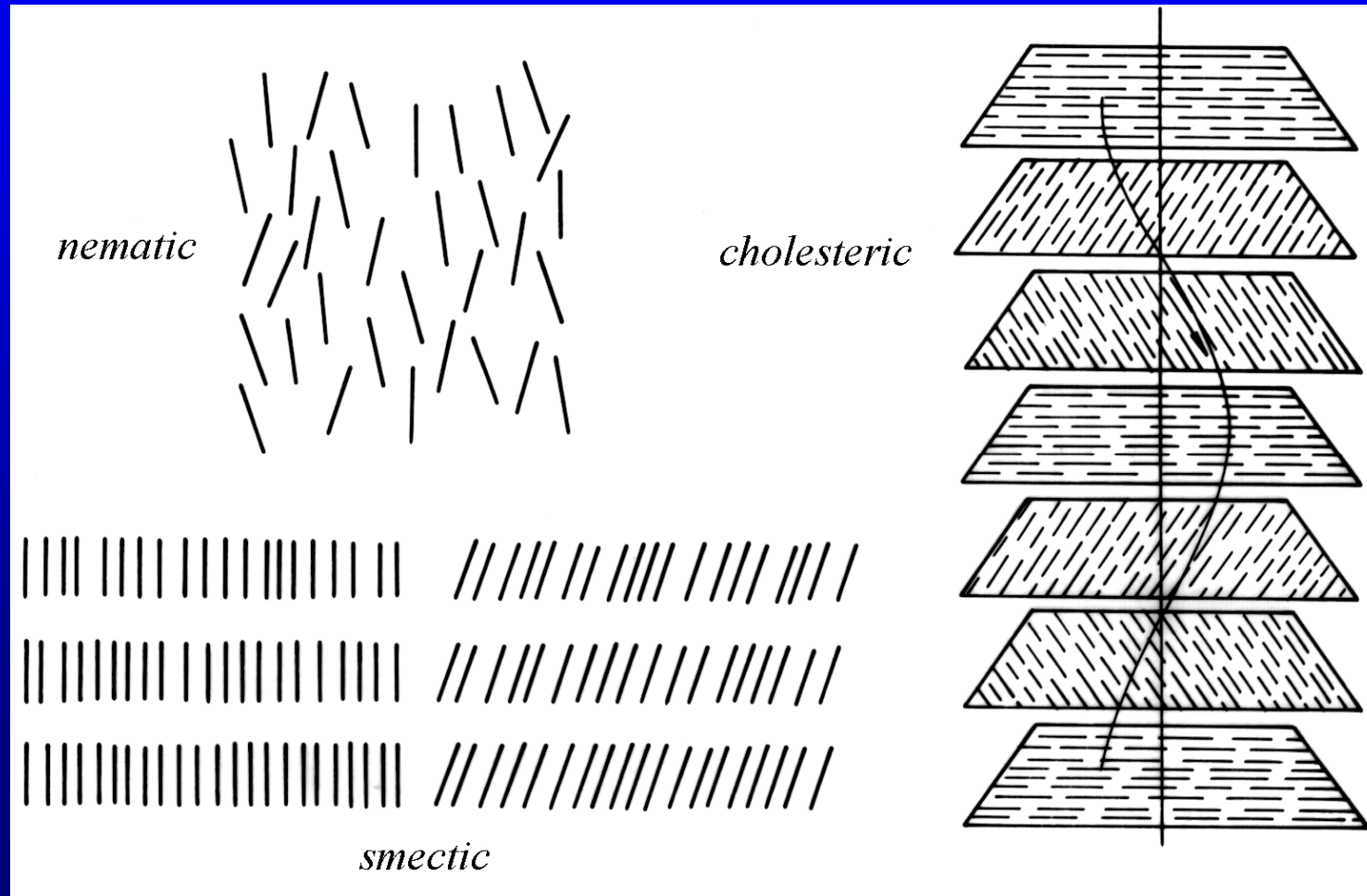
For rod-like molecule of liquid crystal

Nematic phase (向列相)

orientation order

cholesteric phase (胆甾相)

smectic (近晶相)



solids of (magnetic or electric) polar atoms (ions, molecules) may assume magnetic or electric orders

Quasi-crystals are a new class of ordered structures.

They do not have translational symmetry.

Instead they have **bond orientation orders** alone, some being self-similar

The discovery of quasicrystals :
math, computer simulation and experiments

1D quasicrystal: arranged according to Fibonacci
sequence

	sequence	Fibonacci number
F_0	S	1
F_1	L	1
F_2	LS	2
F_3	LSL	3
F_4	LSLLS	5
F_5	LSLLSLSL	8
F_n	$F_{n-1} + F_{n-2}$	

S~short bond; L~long bond→ quasi-crystal

Fibonacci series

$$F_1 = 1; F_2 = 1; F_n = F_{n-1} + F_{n-2}$$

Each member of the series is the sum of the previous two

1, 1, 2, 3, 5, 8, 13, 21, 34, 55, 89, 144, 233, 377, 610, 987, 1597, 2584, 4181, 6765, 10946, 17711, 28657, 46368, 75025, 121393, 196418, 317811, 514229, 832040, 1346269, 2178309, 3524578, 5702887, 9227465, 14930352, 24157817, 39088169, 63245986, 102334155, ...

The ratio of consecutive terms, F_{n+1}/F_n , tends to the golden ratio, $(1 + \sqrt{5})/2 = 1.61803...$

$$\frac{L+S}{L} = \frac{L}{S}, \quad \frac{L+S+L}{L+S} = \frac{L+S}{L}, \dots$$

Diagram illustrating the construction of a Fibonacci sequence from L and S segments. The top row shows L and S segments. The middle row shows the sequence L, S, L+S, L+S+L, L+S+L+L, etc. The bottom row shows the sequence of boxes: [L S] [L] [L+S] [L+S+L] [L+S+L+L], etc.

Self-similarity requires that

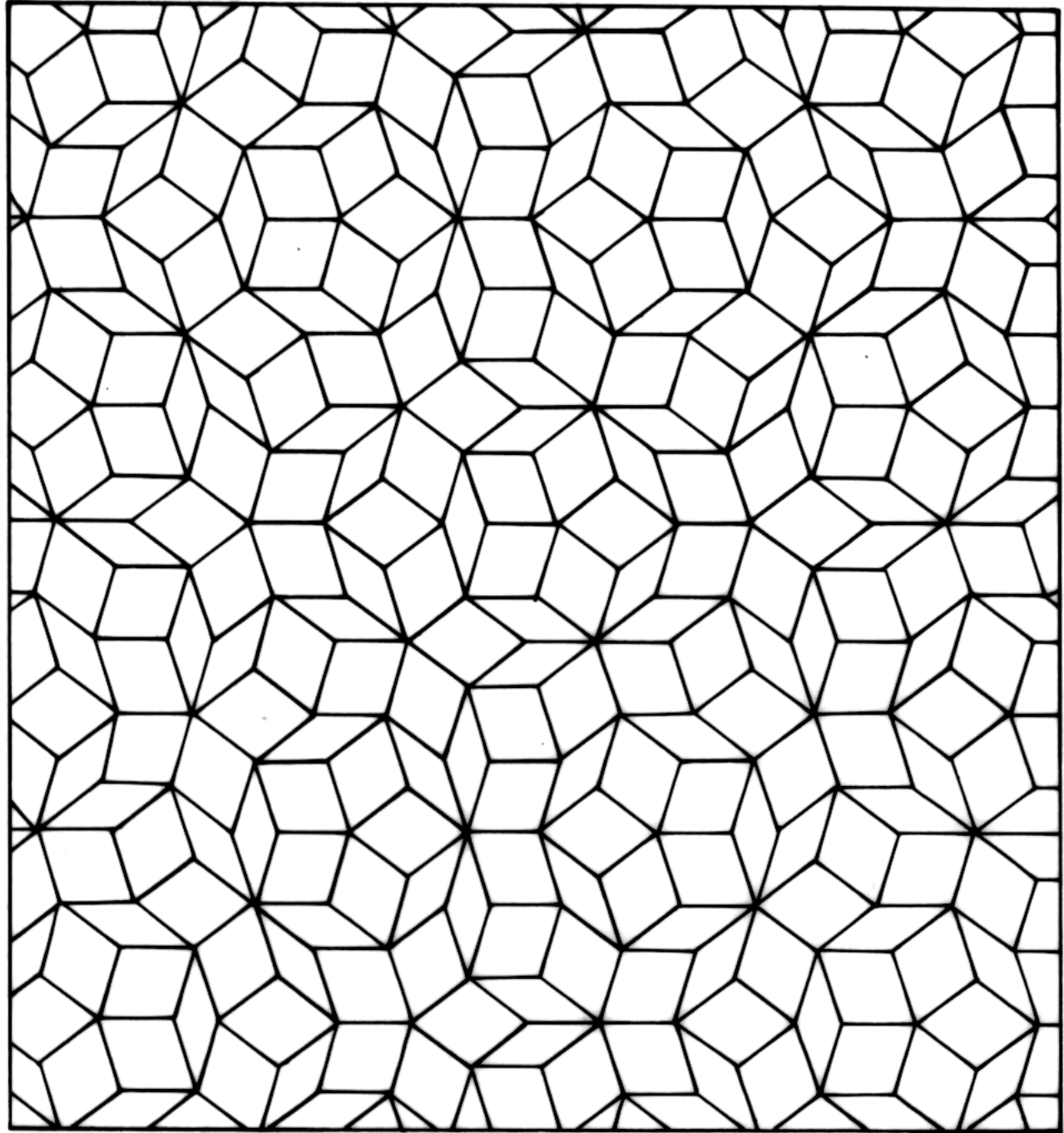
$$\frac{L+S}{L} = \frac{L}{S}, \quad \left(\frac{L}{S}\right)^2 - \frac{L}{S} - 1 = 0.$$

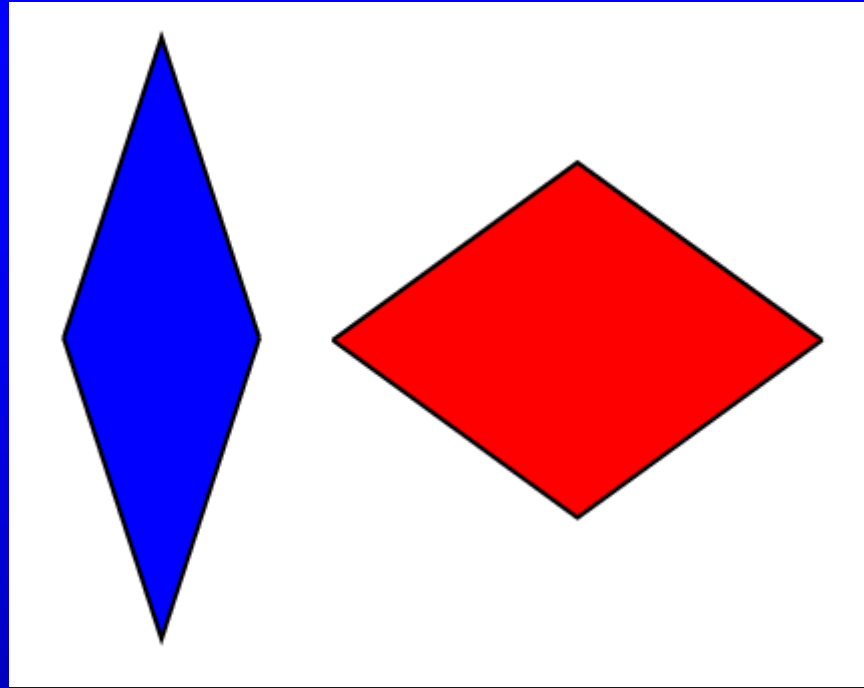
$$\frac{L}{S} = \tau = \frac{1+\sqrt{5}}{2}. \quad \text{the golden ratio}$$

Several aperiodic tilings discovered by the British mathematician/physicist Roger Penrose

A way of precisely covering an infinite plane with a fixed set of shapes.

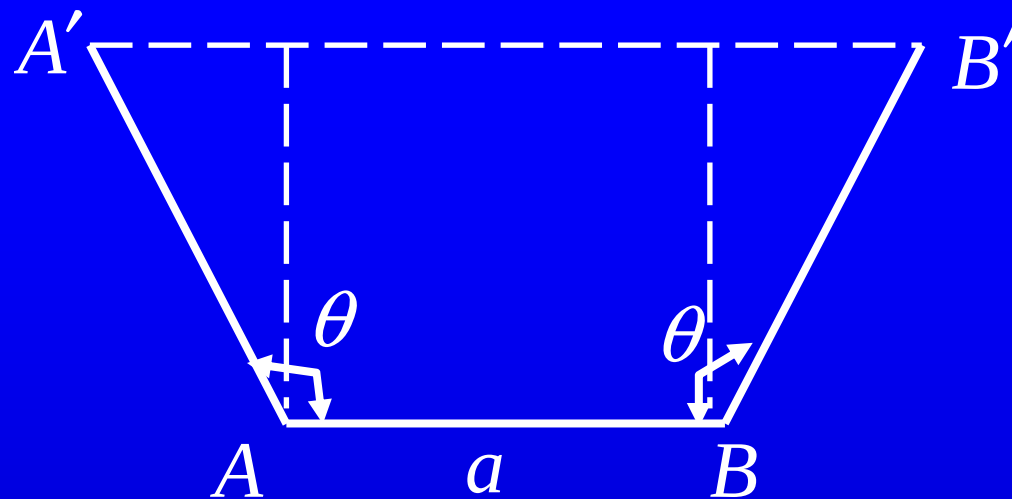
Penrose
tiling.
Penrose
packing.
& five-
fold
symmetry





Penrose rhombs (rhomb)

36- and 72-degree rhombi (rhombus)



$$A'B' = n \cdot AB, \quad n \in \mathbb{Z}$$

$$A'B' = AB + 2AB \cos(\pi - \theta) = AB - 2AB \cos(\theta)$$

$$n = 1 - 2 \cos(\theta)$$

$$\because -1 \leq \cos(\theta) \leq 1$$

$$\therefore n = -1, 0, 1, 2, 3$$

$$\therefore \theta = 2\pi, \frac{2\pi}{2}, \frac{2\pi}{3}, \frac{2\pi}{4}, \frac{2\pi}{6}$$

**Only 1-, 2-, 3-, 4- and 6- fold rotation axes
consistent with translational symmetry.**

quilt



Quasi-periodicity

periods

a b

$a/b = \text{rational number} = m/n$

$na = mb$ the common period

$a/b = \text{irrational number}$ no common period

incommensurate

$$f(x) = \cos x + \cos \pi x$$

period 2π 2

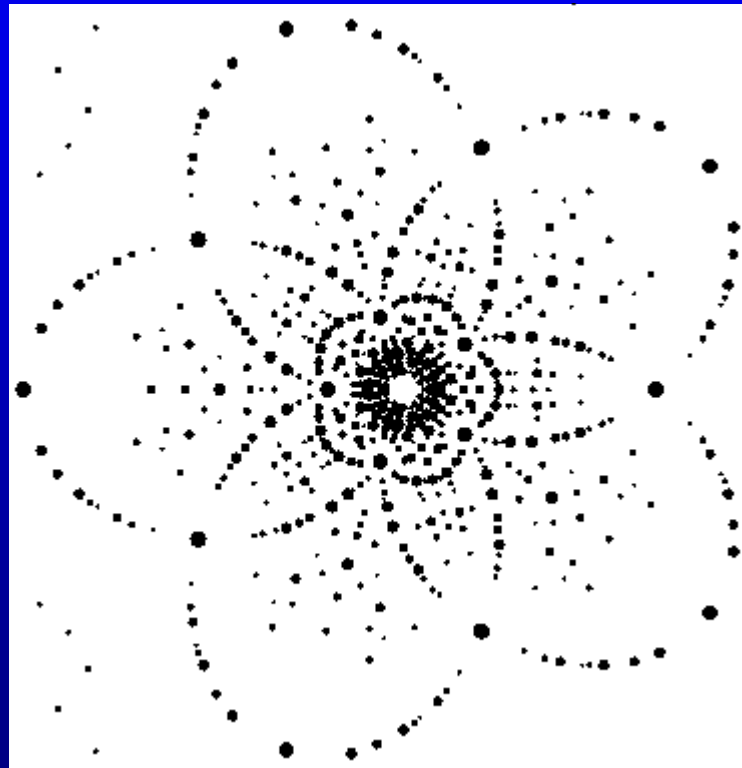
1984 Dan Shechtman

quenched aluminium manganese alloy

quench 淬火 anneal 退火

icosahedral (20面体) packing

a simulation for a Laue pattern (X-ray) from
an icosahedral quasicrystal, whereby the x-
ray beam is along one of the five-fold axes

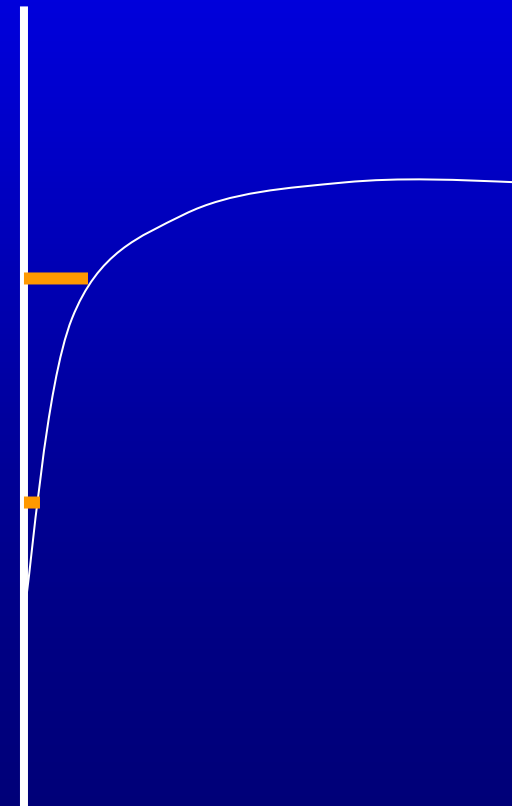
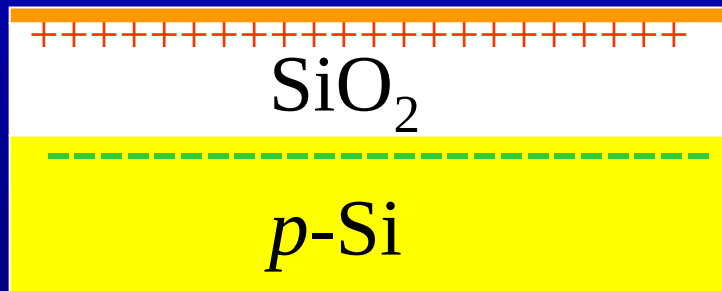


28.2 Fractals

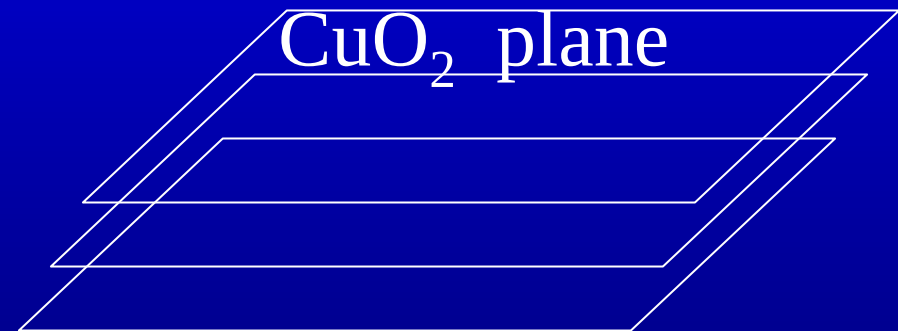
Low dimensional system

- 2D electron gas, the inversion layer in semiconductor

MOSFET $+V$



- Semiconductor heterostructure
- the electron crystal on the liquid He surface —Wigner solid
- quantum well
- high temperature superconductor



1D objects

- quantum lines
- polymers

0D objects

- quantum dots
- superfine particles

nanostructure

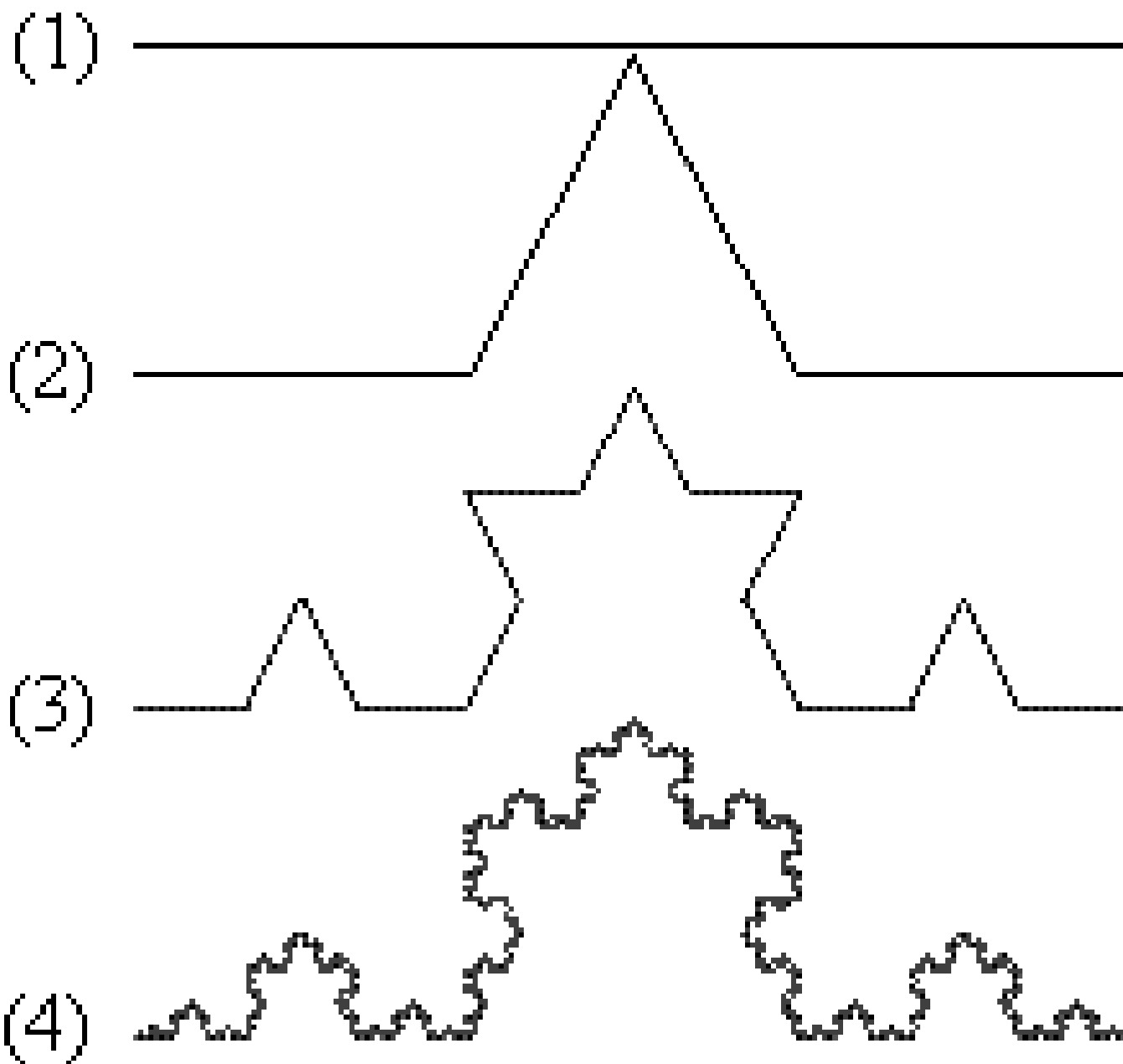
macroscopic

mesoscopic

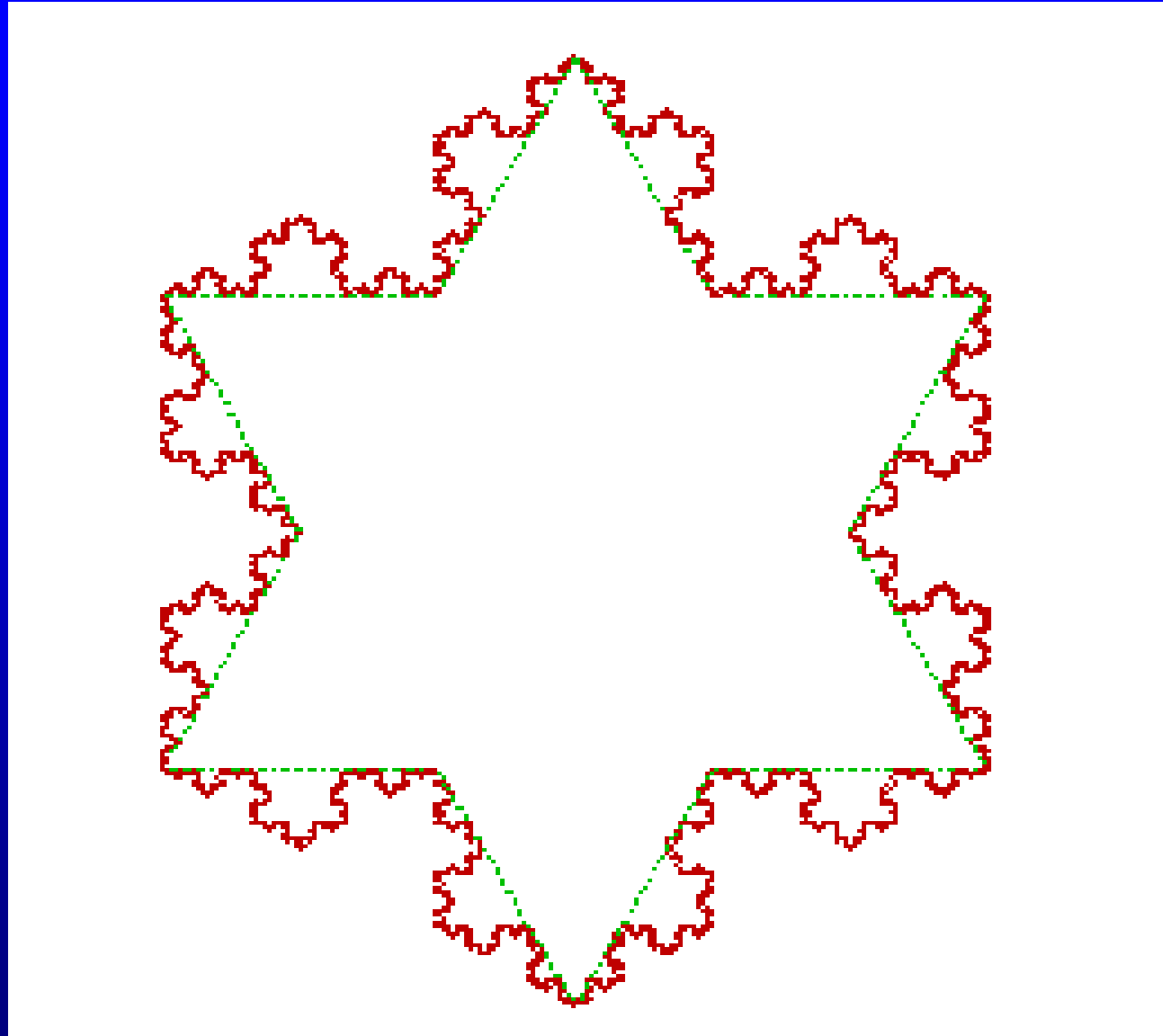
microscopic

Koch
curve

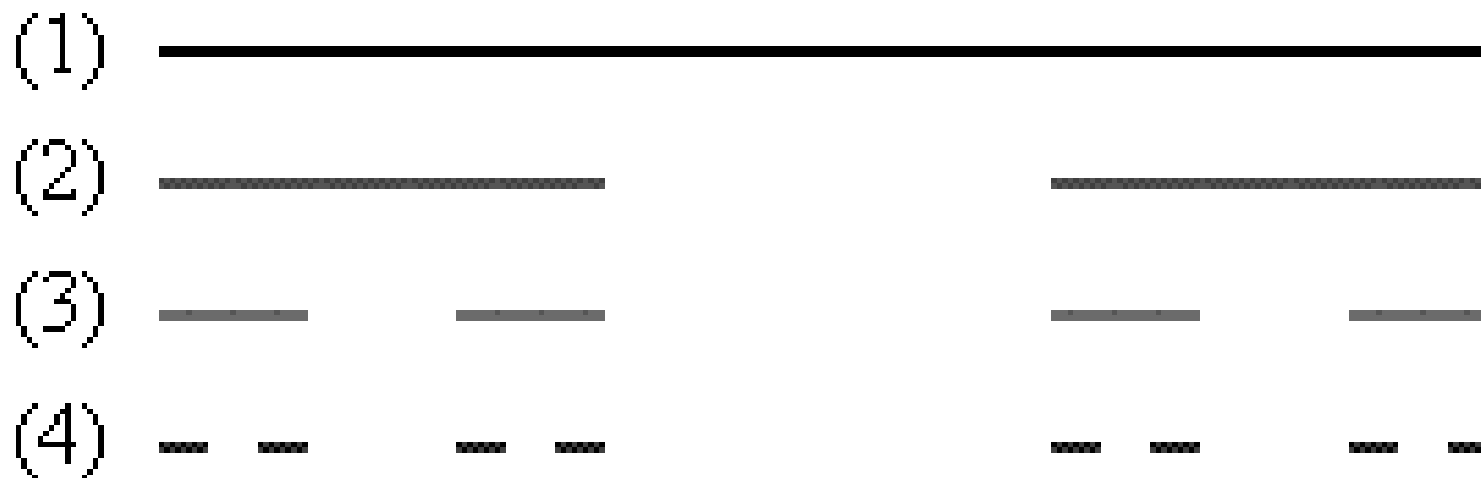
$d \sim 1-2$?



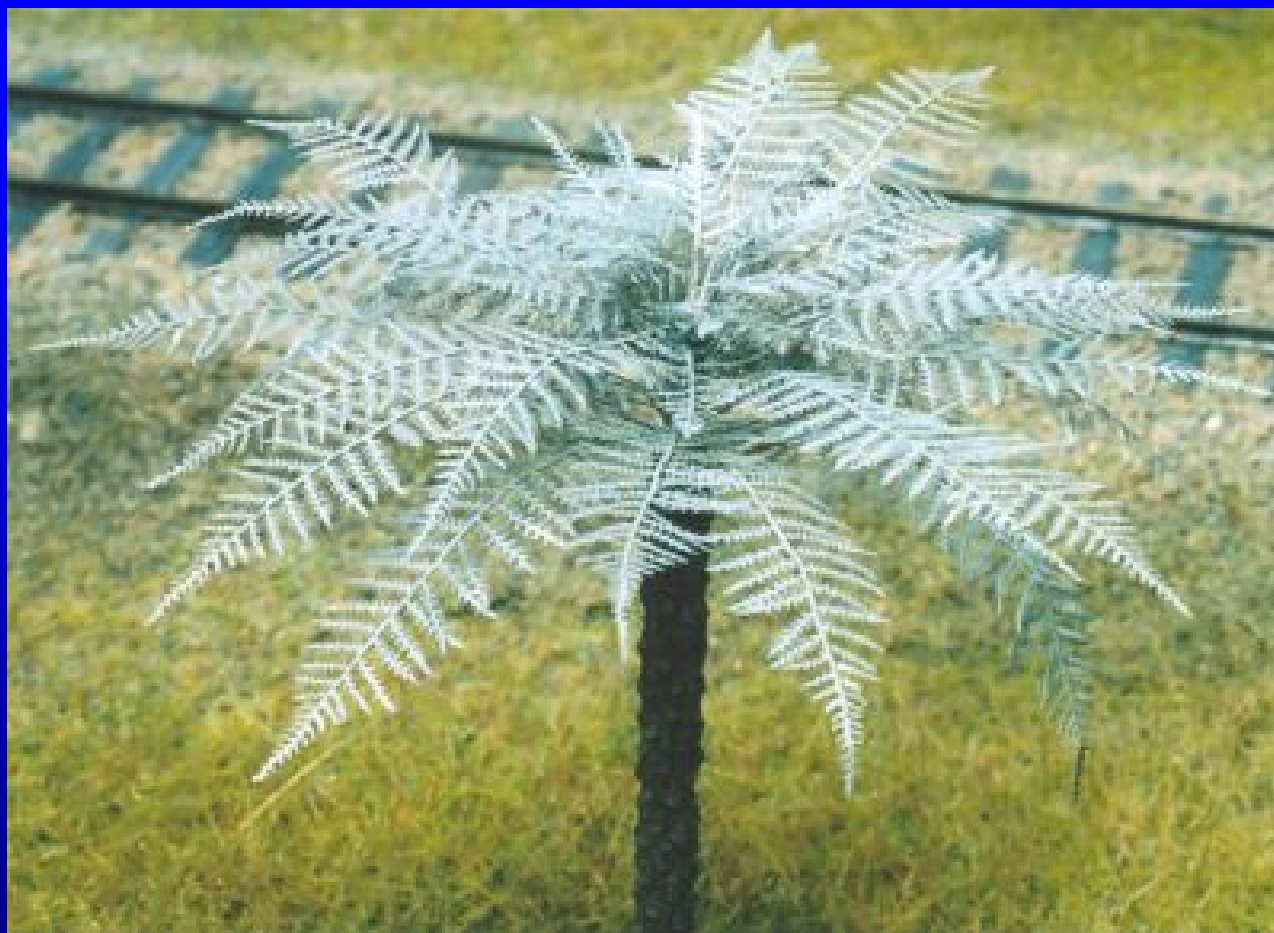
Koch
snowflake



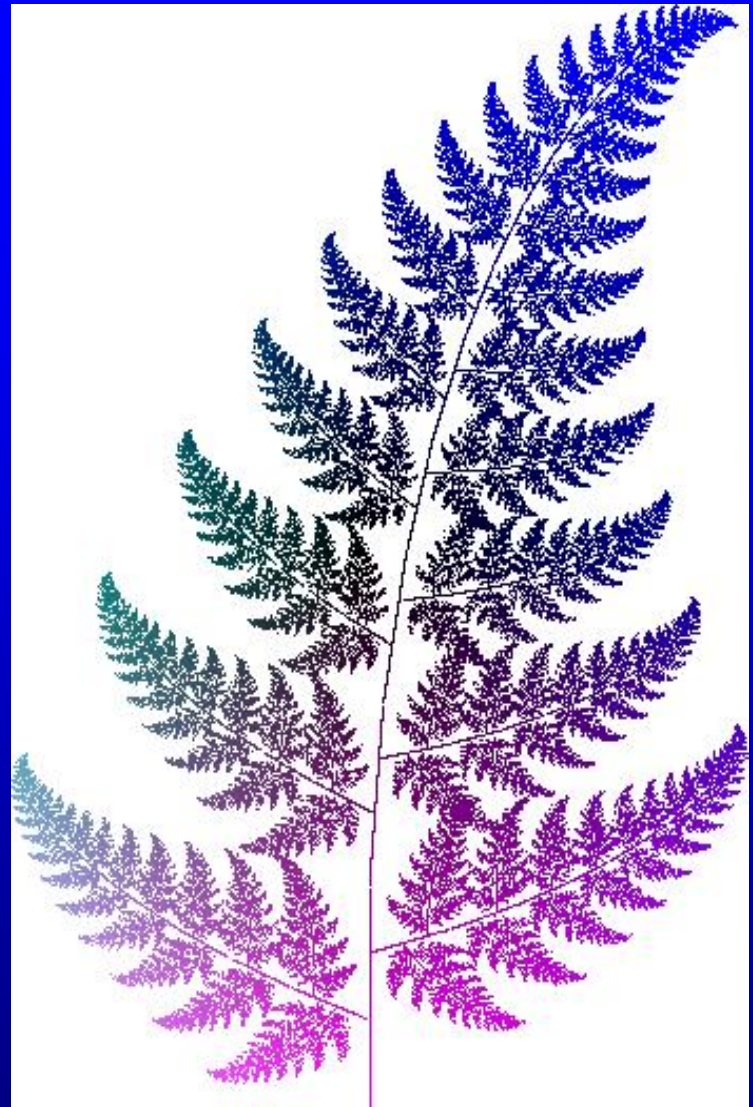
Cantor set



$d \sim 0 - 1?$

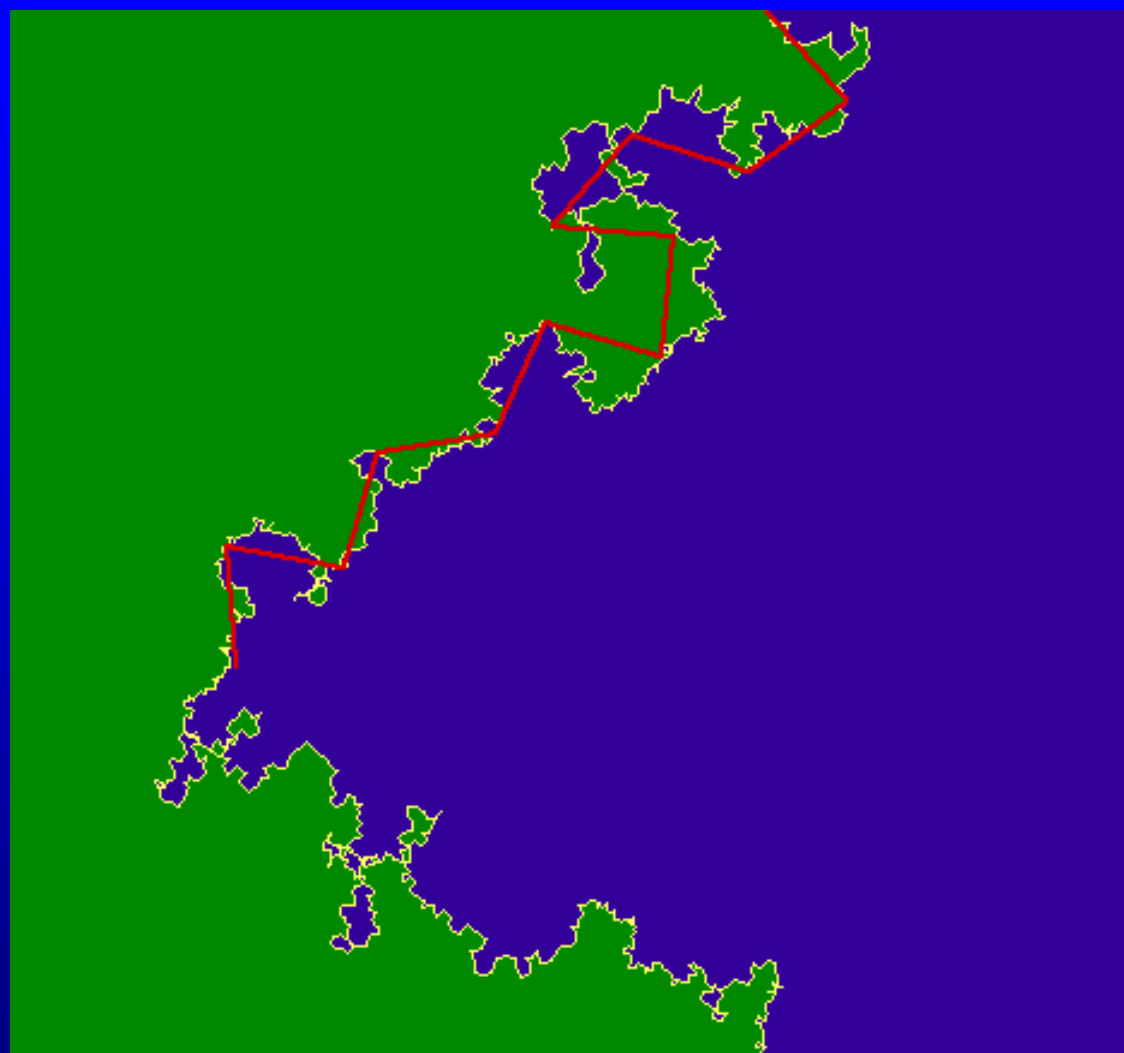


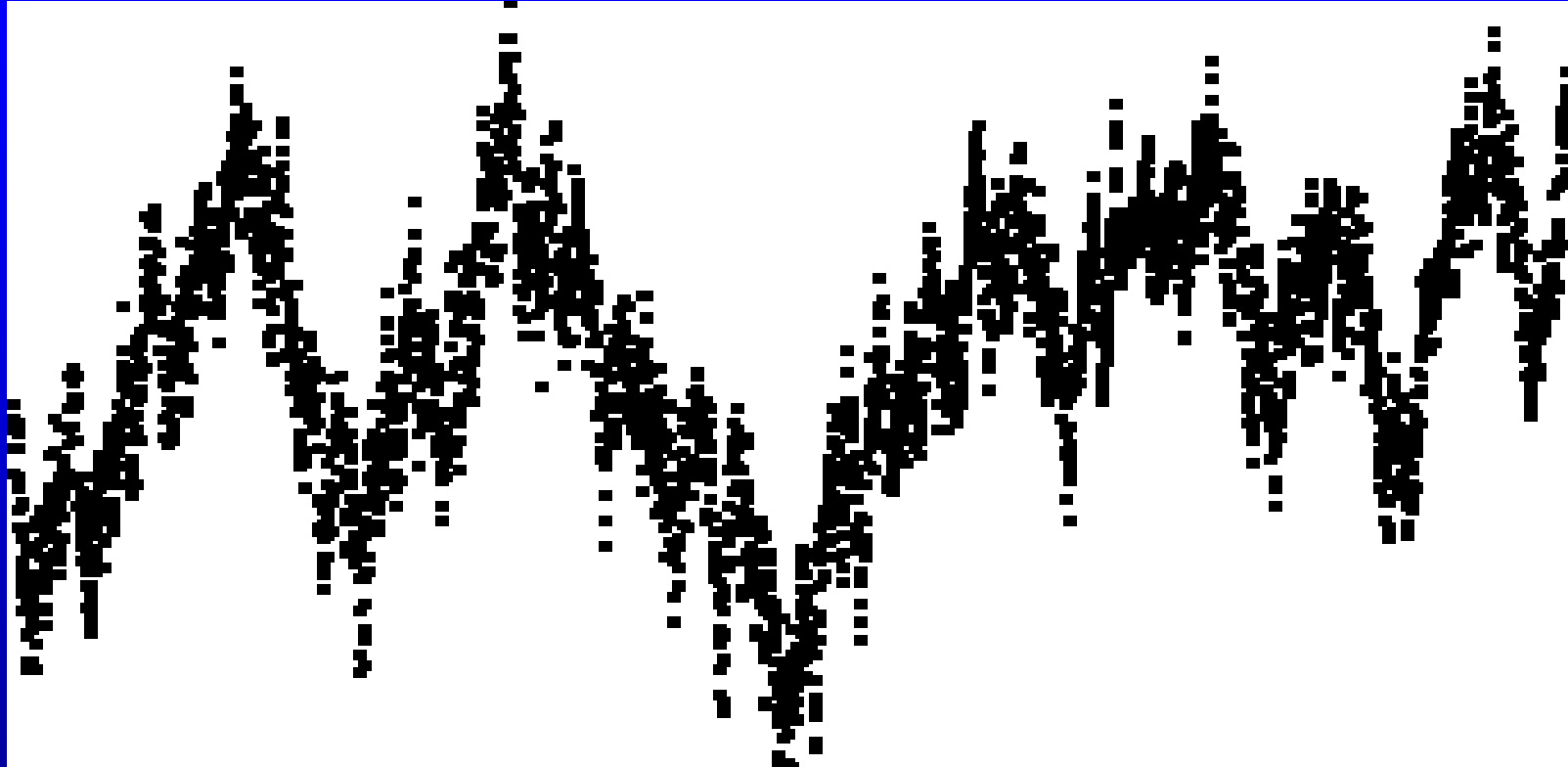
fern 蕨树



foliage of fern





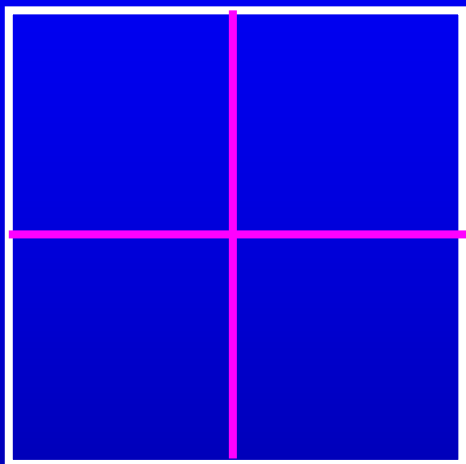


Fractal dimensions



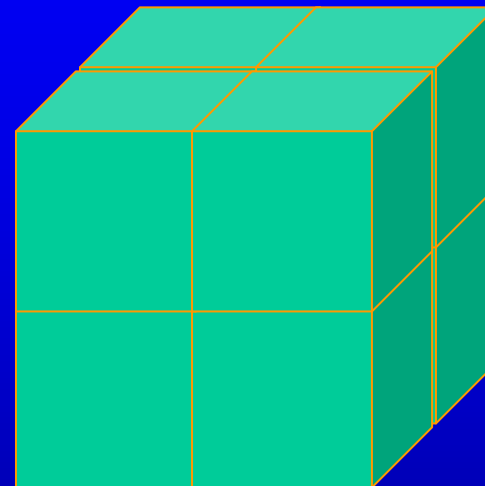
$$2^1$$

$$1 = \frac{\ln 2}{\ln 2}$$



$$2^2$$

$$2 = \frac{\ln 4}{\ln 2}$$



$$2^3$$

$$3 = \frac{\ln 8}{\ln 2}$$

Similarity dimension

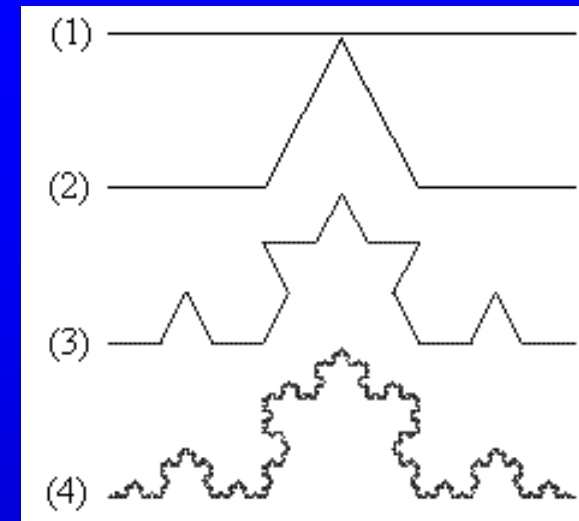
When a shape is composed of n similar shape of size $1/m$, then the similarity dimension is given by

$$D_s = \frac{\ln n}{\ln m}.$$

Koch curve

4 miniatures of size 1/3

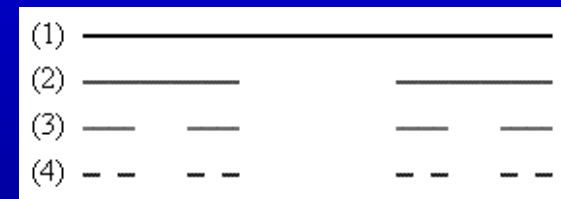
$$D_s = \frac{\ln 4}{\ln 3} = 1.2618\dots$$



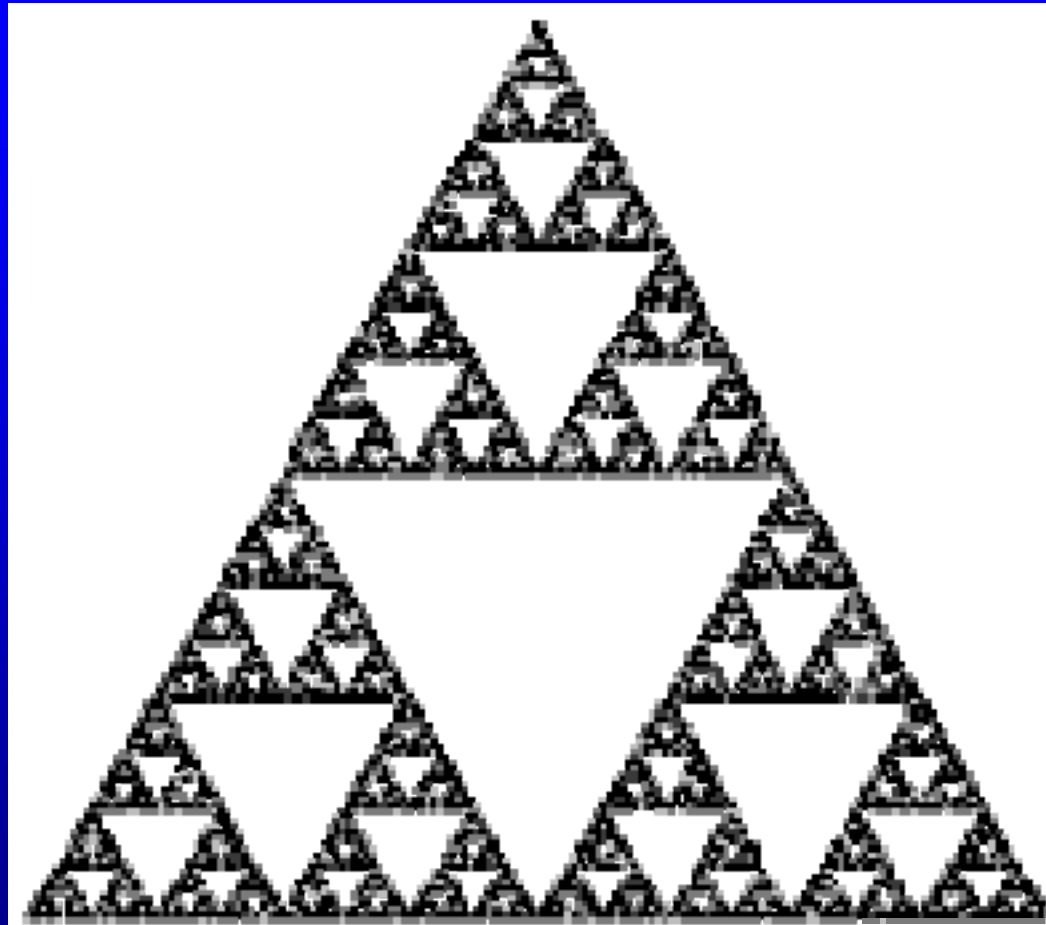
Cantor set

2 miniatures of size 1/3

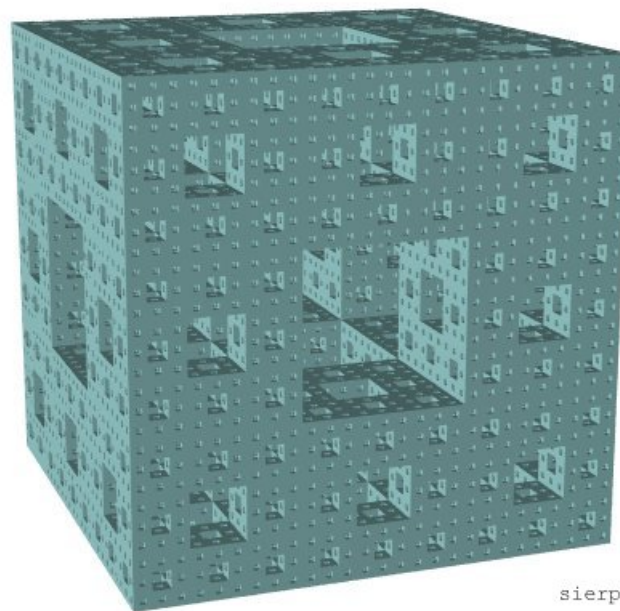
$$D_s = \frac{\ln 2}{\ln 3} = 0.63\dots$$



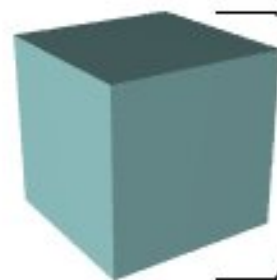
The Sierpinski gasket



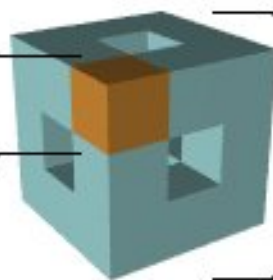
The Sierpinski sponge



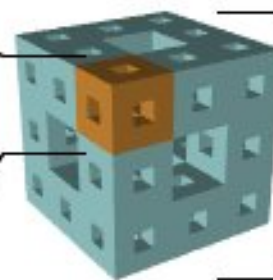
sierpinski 4



sierpinski 0



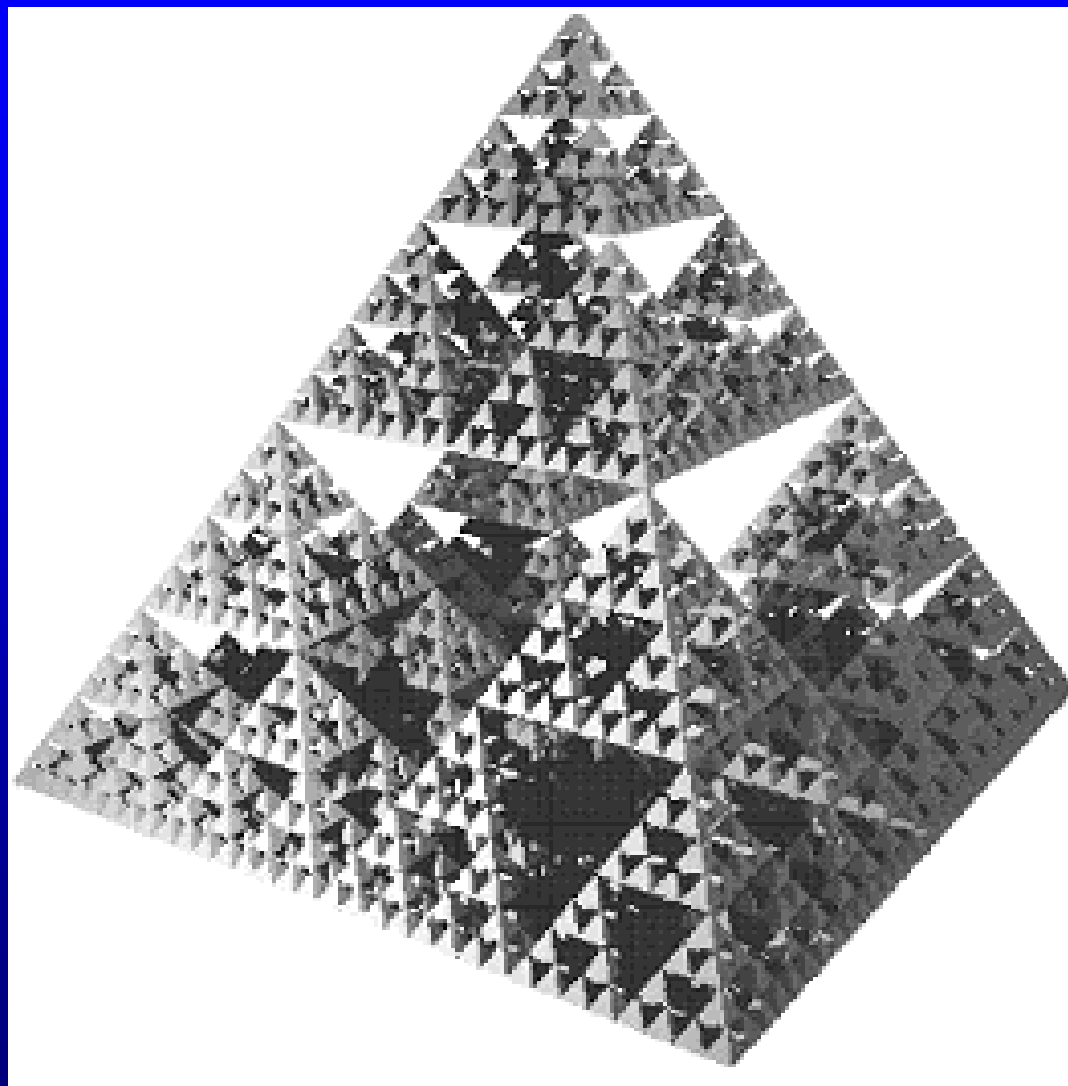
sierpinski 1



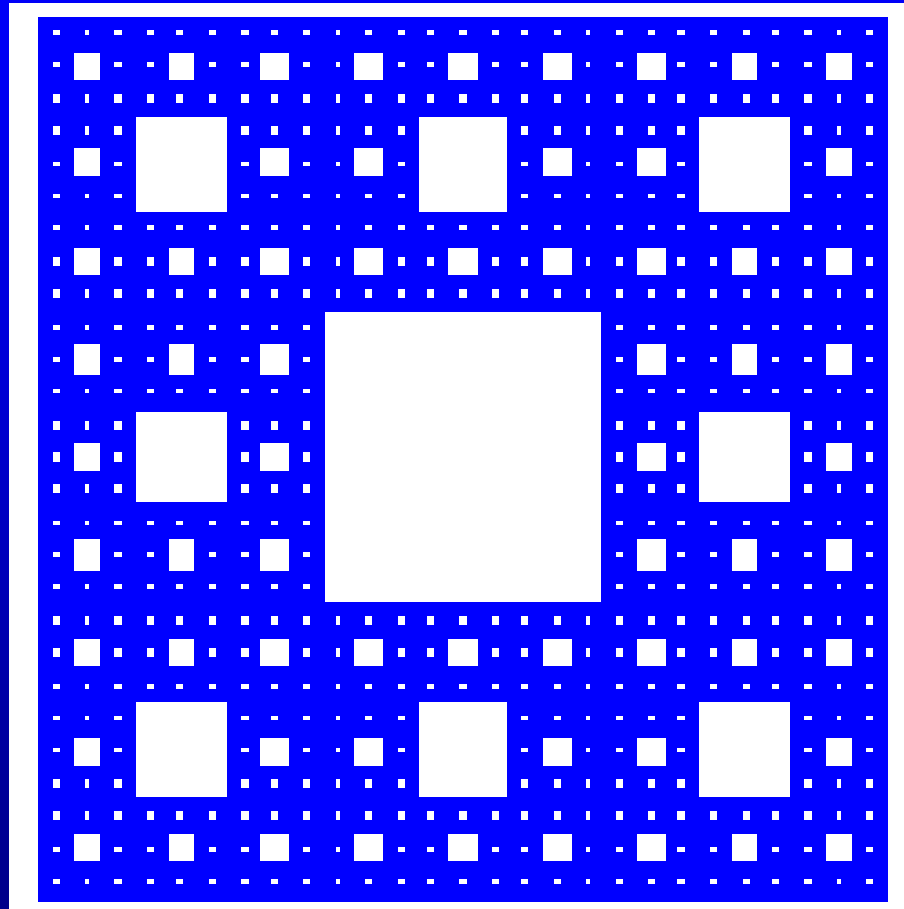
sierpinski 2

O.S.V.

The Sierpinski gasket in 3 dimensions



The Sierpinski carpet



Capacity dimension

Consider a bounded set in d -dimensional Euclidean space. Cover the set by d -dimensional spheres of identical radius ε . The capacity dimension is

$$D_c = \lim_{\varepsilon \rightarrow 0} \frac{\ln N(\varepsilon)}{\ln \frac{1}{\varepsilon}}$$

where $N(\varepsilon)$ denotes the minimal number of spheres.

Hausdorff dimension

measure

Koch curve 1D measure (length) infinite

 2D measure (area) zero

$$D = \frac{\ln 4}{\ln 3} = 1.2618\dots \quad \text{the measure is finite}$$

Most coastlines have non-integer Hausdorff dimension.

percolation

polymers

coarse interfaces

metal crackles, and various amorphous materials,

dielectric breakdown

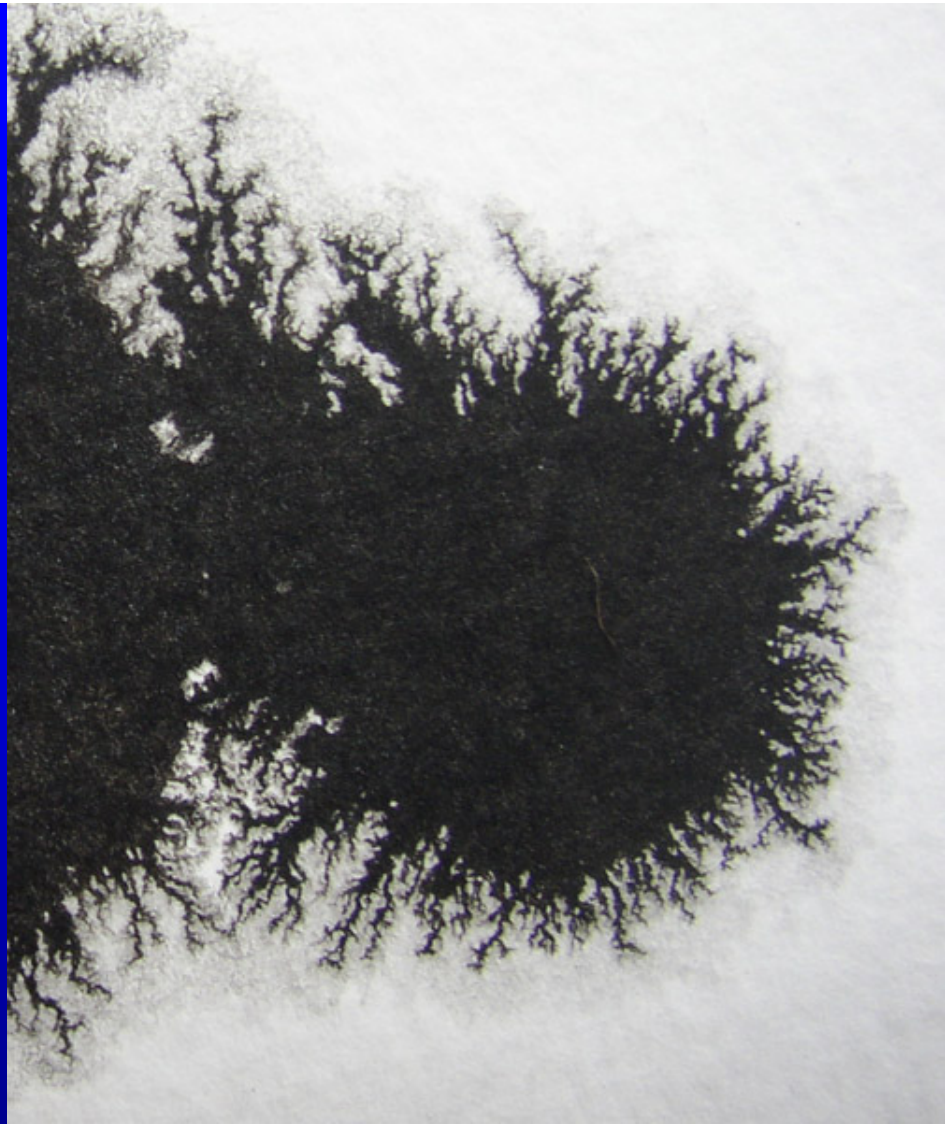
electrochemical deposit

viscous fingers

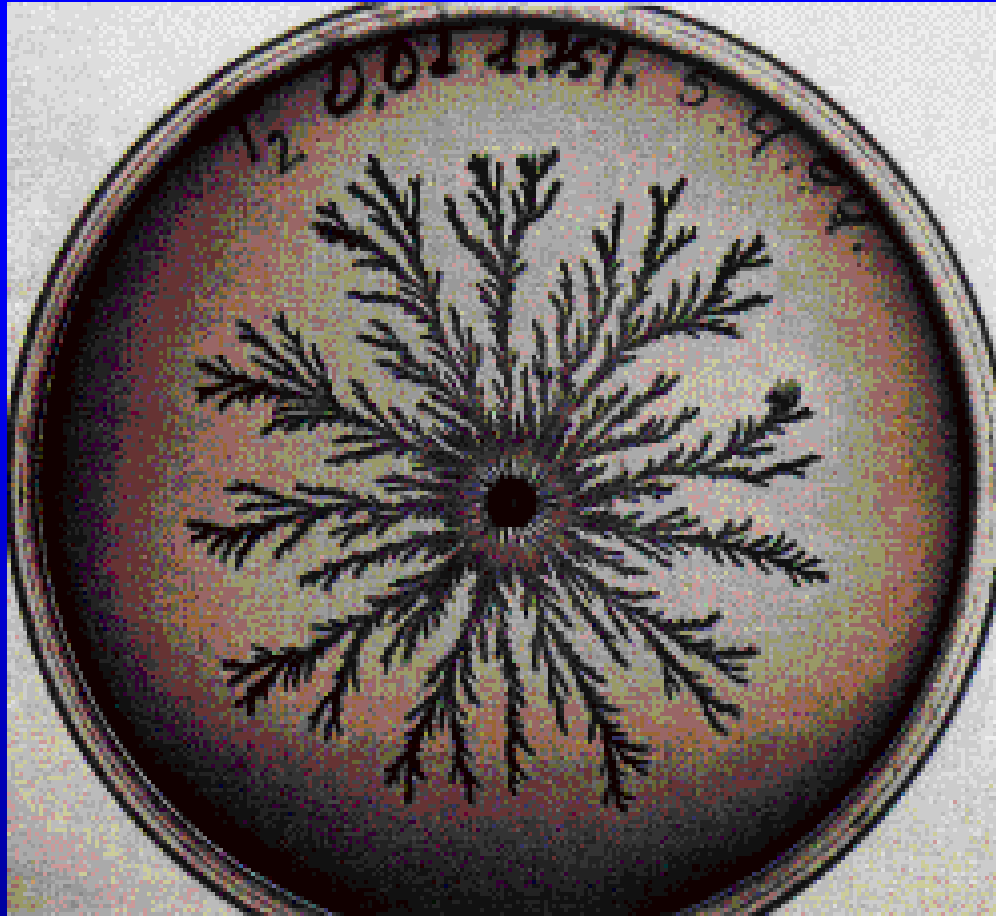
arborescent crystal solidification

fractal geography

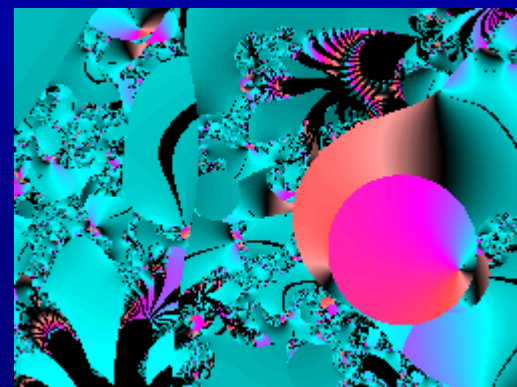
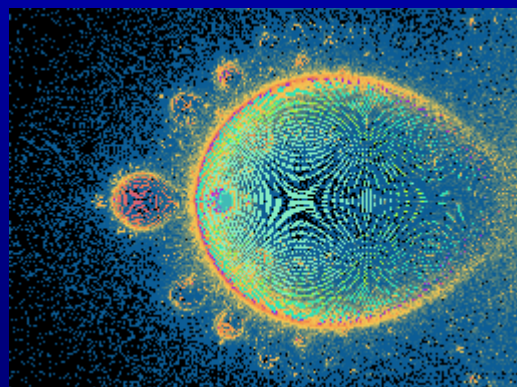
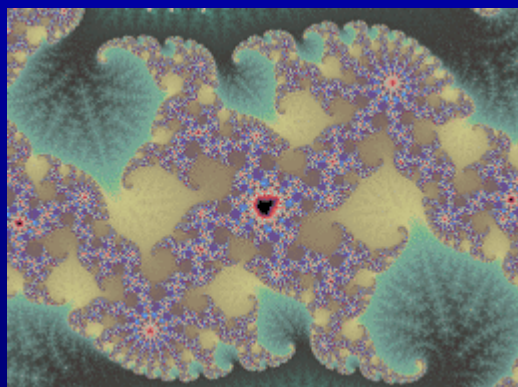
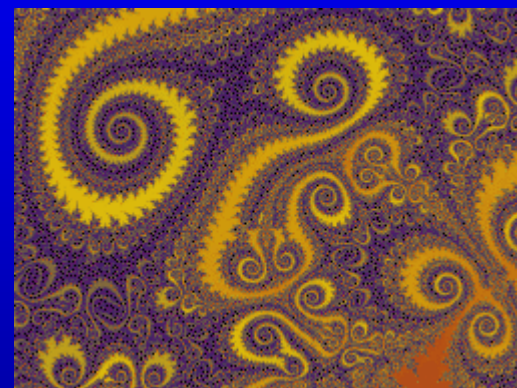
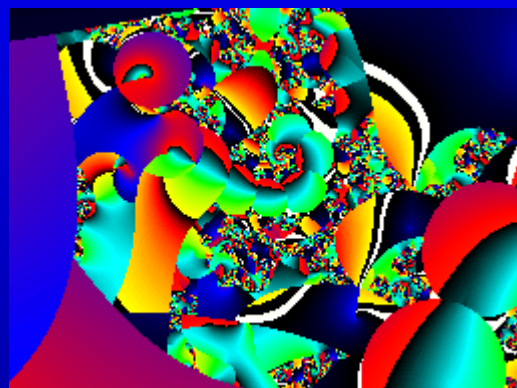
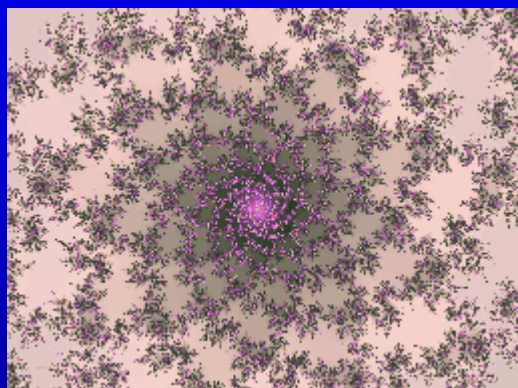
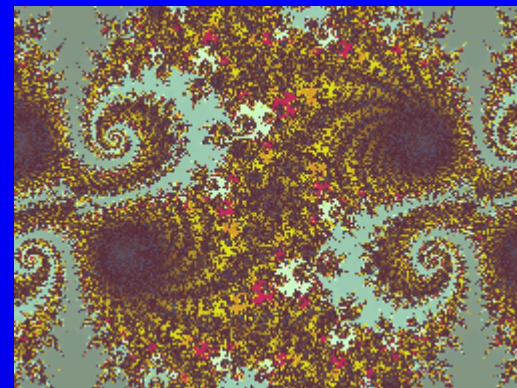
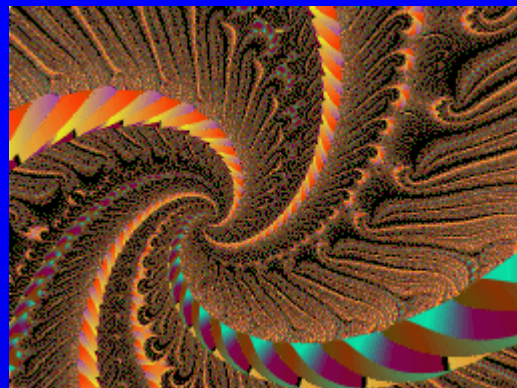
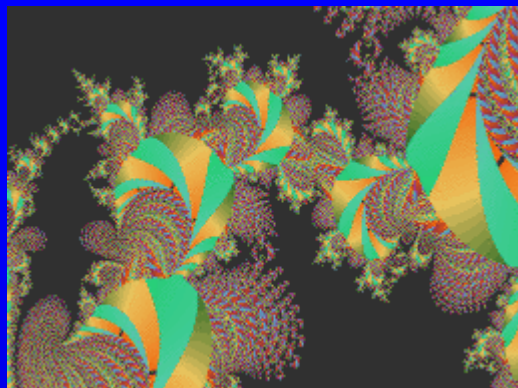
approximate self-similarity

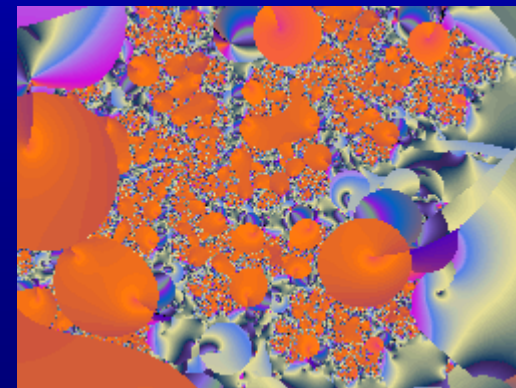
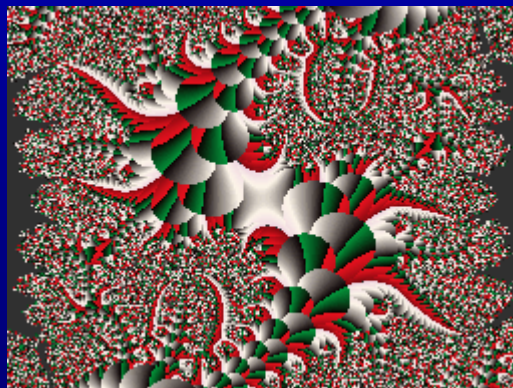
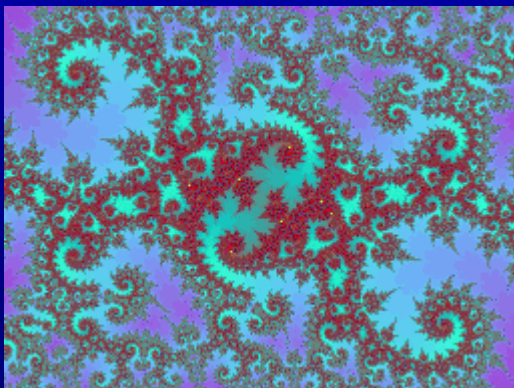
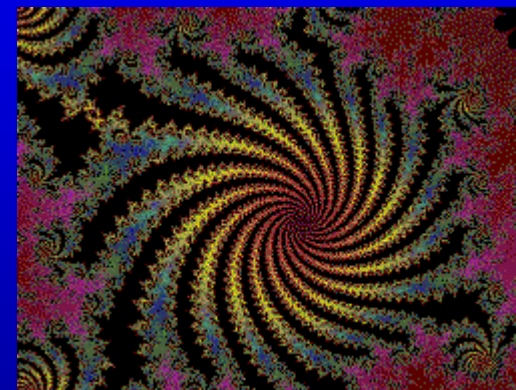
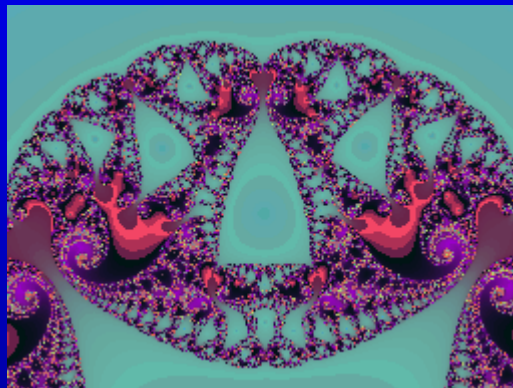
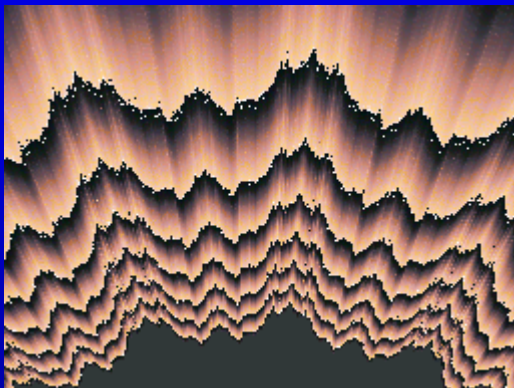
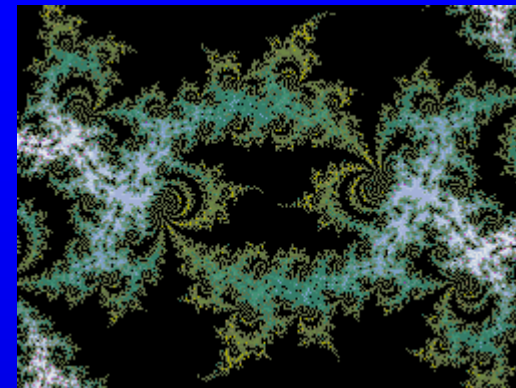
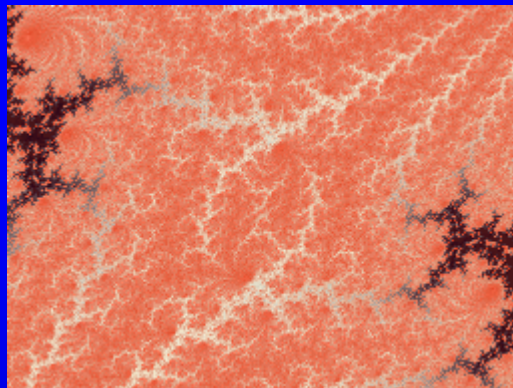
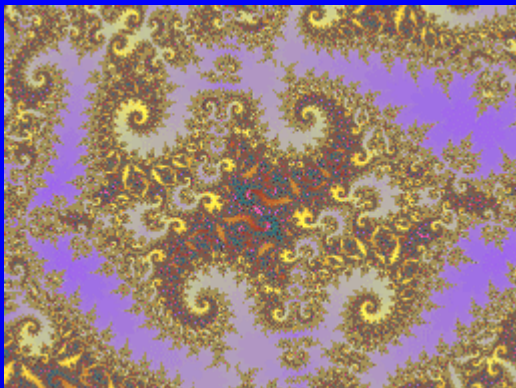


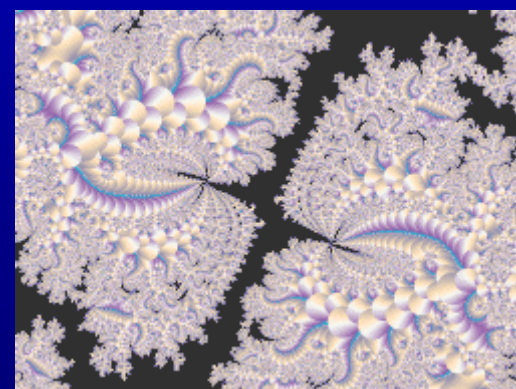
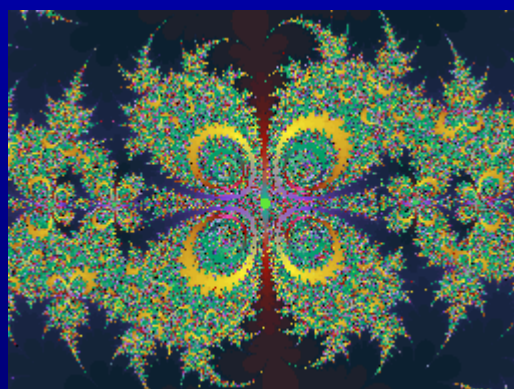
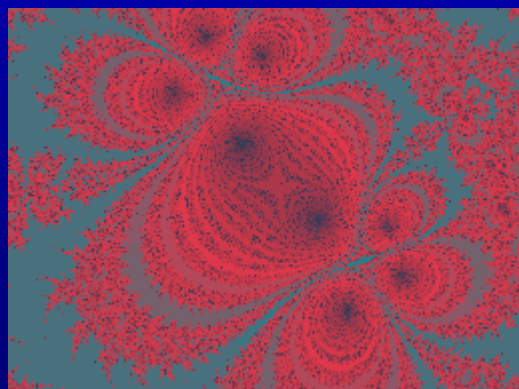
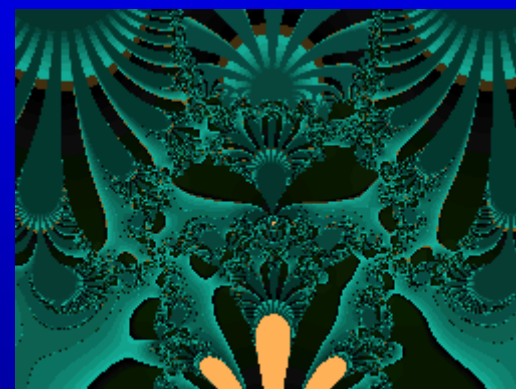
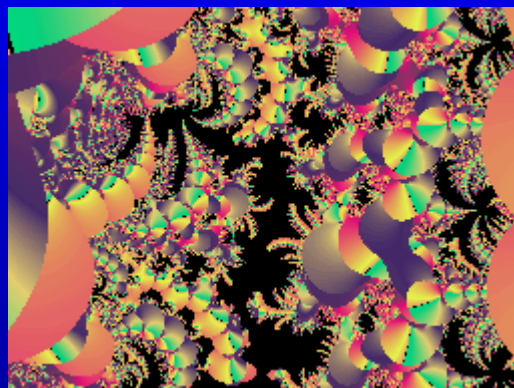
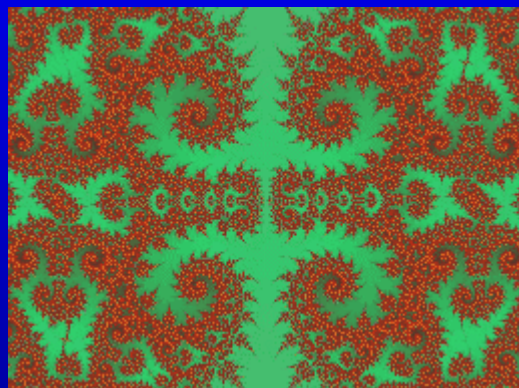
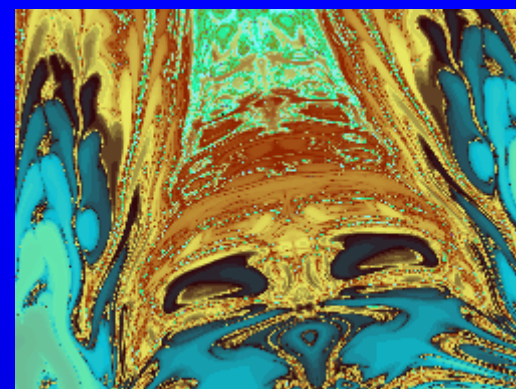
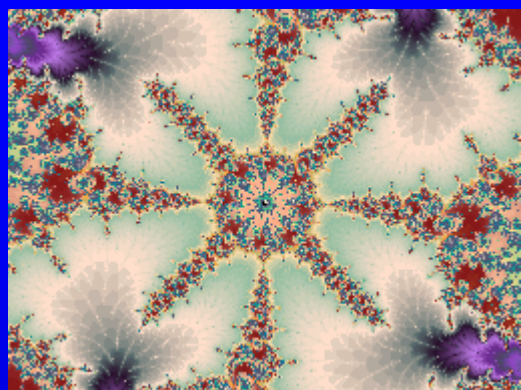
One drop of ink placed in still water results in a Viscous Finger Fractal.



Bacterial Growth in Stressed Environments







Assignment: 28.2, 28.3

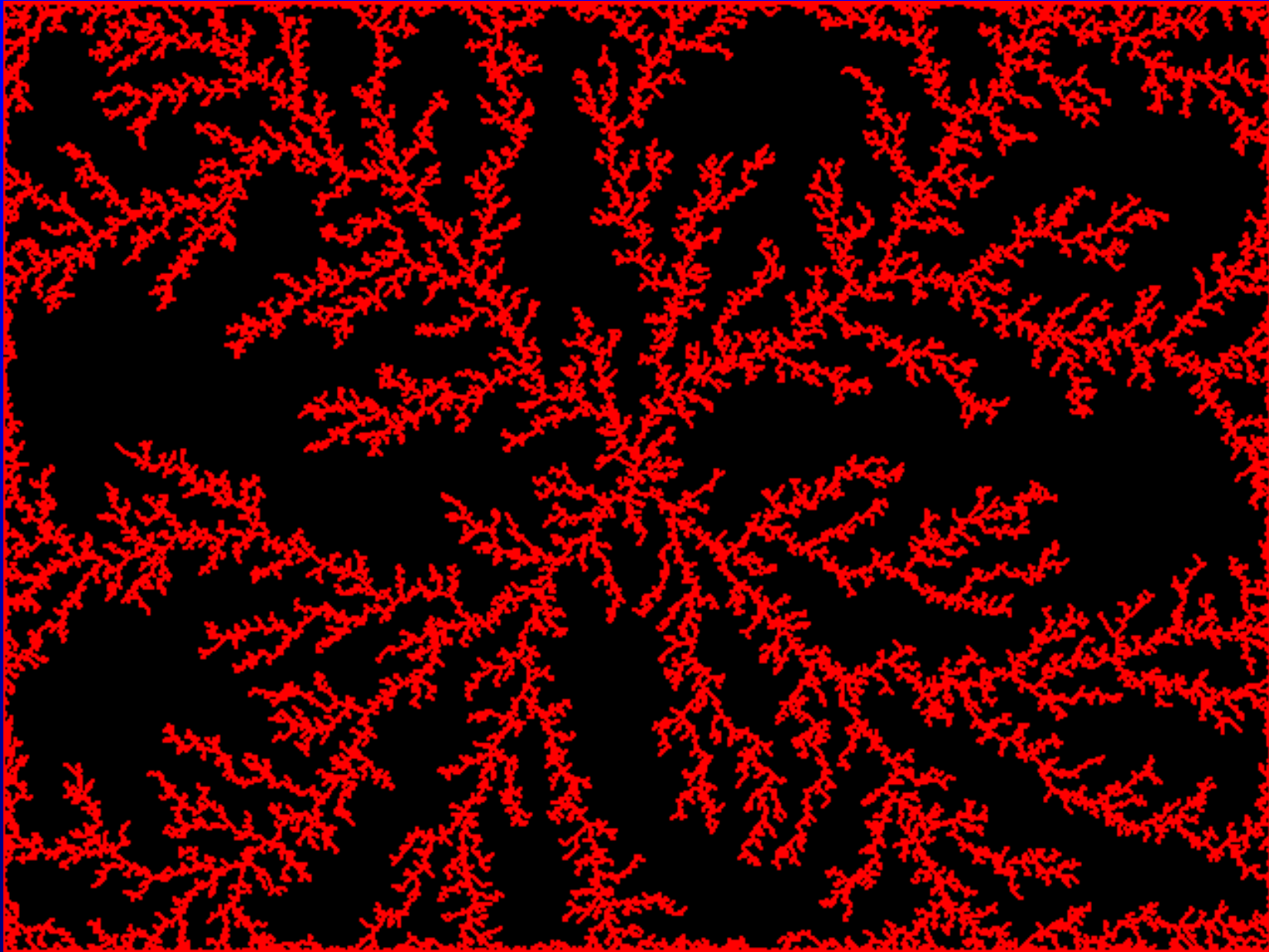
Rongjia Tao Professor of Physics Temple University

Smart Fluids, electrorheological (ER) and magnetorheological (MR) fluids

In research on smart fluids, we have focused on the physical mechanisms, phase transitions, materials, microstructures, and dynamic process. As reported by *Nature* (V358, 373 (1992)), *Scientific American* (October, 58 (1993)), (German) *Rheology* journal (V.3, 284-285, Oct.-Dec. 1993), and *New York Times* (9/24/1996), our research in this area has received great attention and wide interest. On this project, we have extensive cooperation with industries, such as Ford Motor, Lord Corporation, Nippon Shokubai Co. Ltd. (Japan), Asahi Chemical (Japan), and Bridgestone/Firestone. Based on our understanding about the microstructure of MR and ER fluids, we recently invented a novel approach to change the microstructure of these fluids and produced super-strong MR and ER fluids. These fluids are about 10 times stronger than conventional MR and ER fluids. Their impact will be significant.

electrorheological (ER) and magnetorheological (MR) fluids

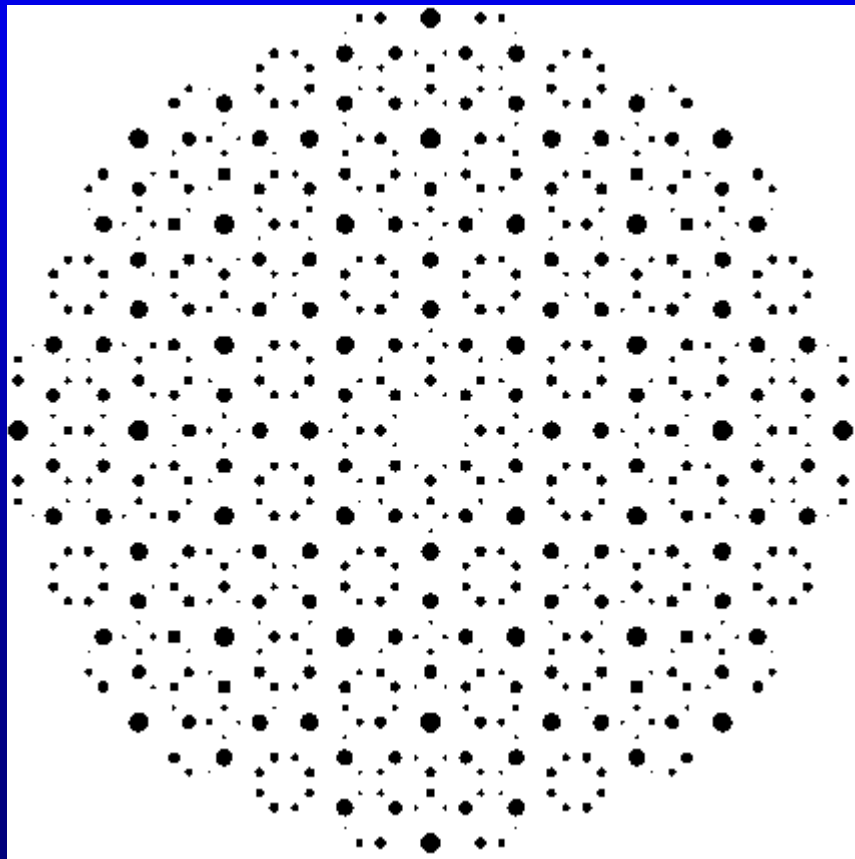
Viscous fingers



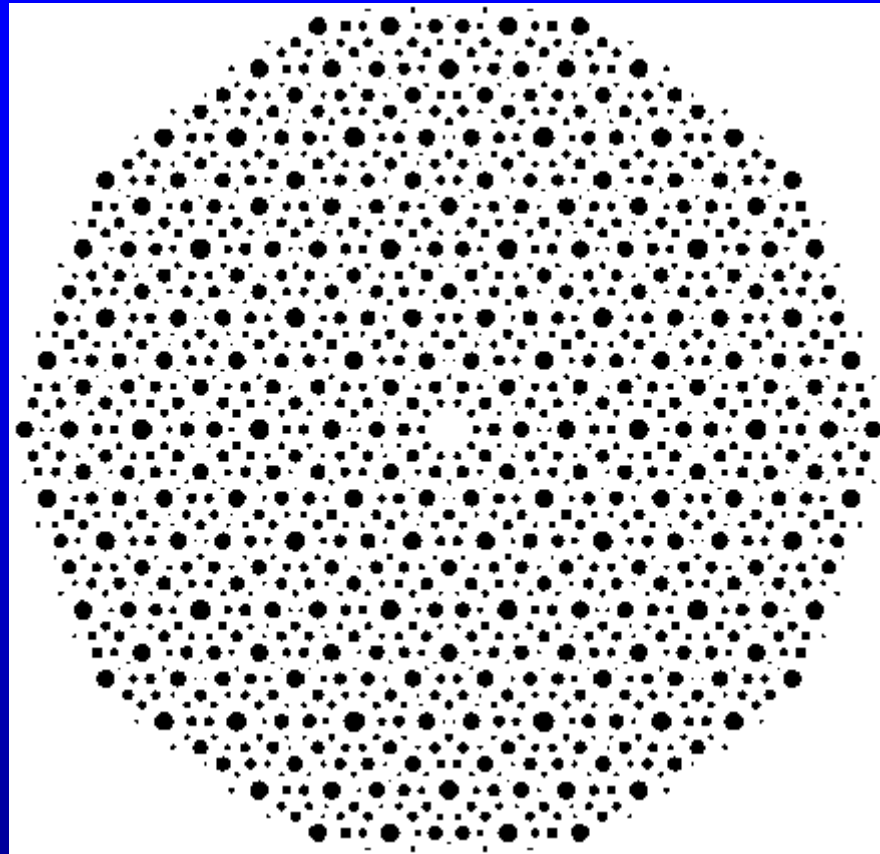
Cellular Automaton (Automata)

A cellular automaton is a collection of "colored" cells on a grid of specified shape that evolves through a number of discrete time steps according to a set of rules based on the states of neighboring cells. The rules are then applied iteratively for as many time steps as desired. von Neumann was one of the first people to consider such a model, and incorporated a cellular model into his "universal constructor." Cellular automata were studied in the early 1950s as a possible model for biological systems (Wolfram 2002, p. 48). Comprehensive studies of cellular automata have been performed by S. Wolfram starting in the 1980s, and Wolfram's fundamental research in the field culminated in the publication of his book *A New Kind of Science* (Wolfram 2002) in which Wolfram presents a gigantic collection of results concerning automata, among which are a number of groundbreaking new discoveries.

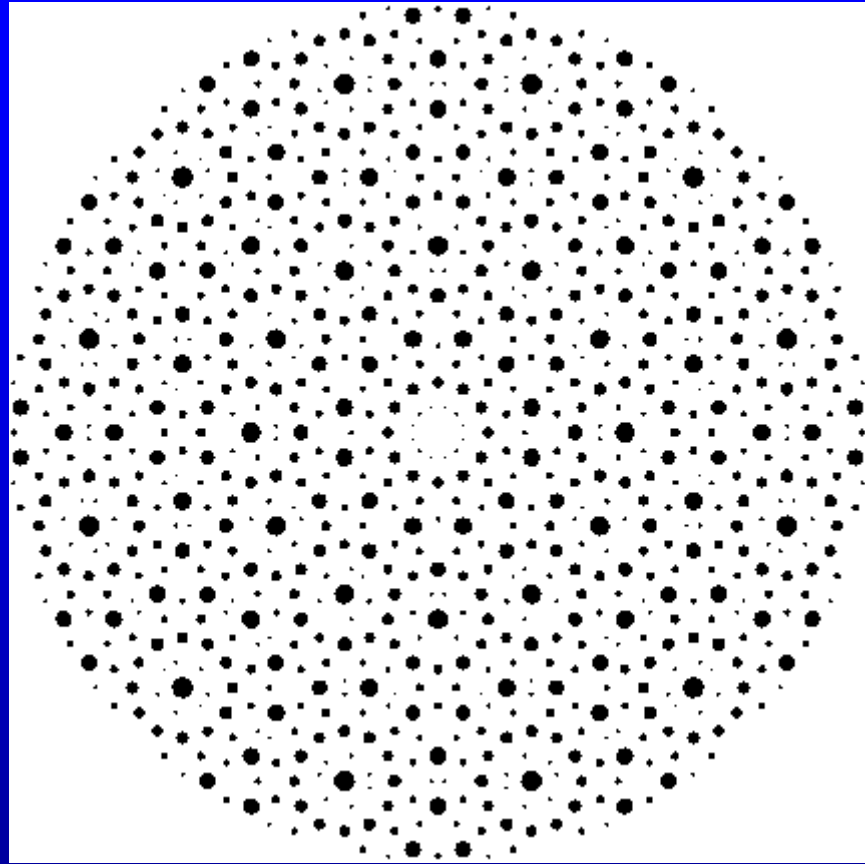
some simulations of diffraction patterns are shown, which could represent either electron diffraction patterns or the zeroth layers of precession photographs (X-ray):



Octagonal(八角形的)QC



decagonal (十边形的)QC



Dodecagonal(十二角形的) QC

28.3 Bonds in crystals

Ionic solids

NaCl ionic crystal

Na^+

closed-shell ions

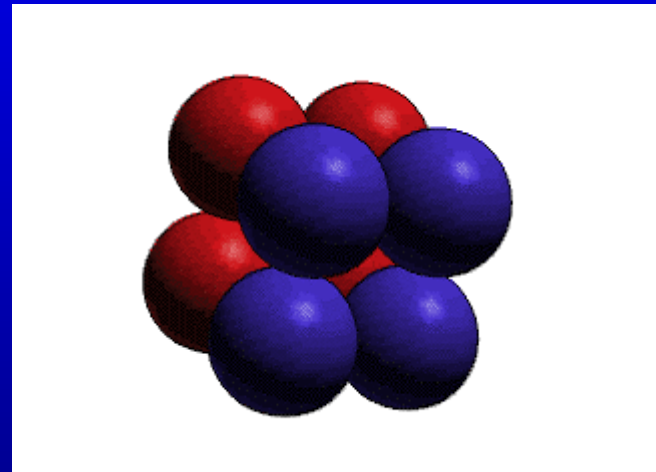
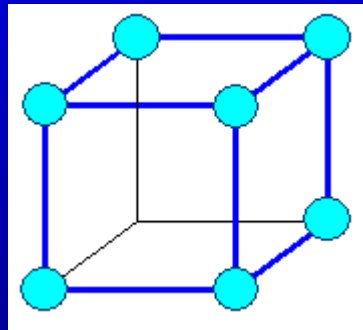
Cl^-

A positive ion can attract many negative ions simultaneously. $\longrightarrow$ solid structure.

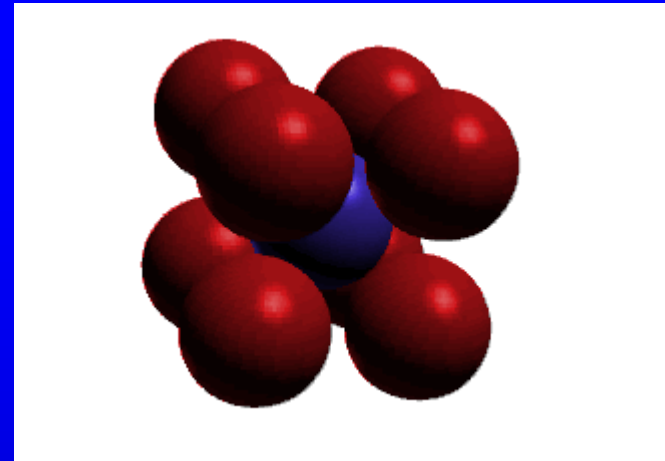
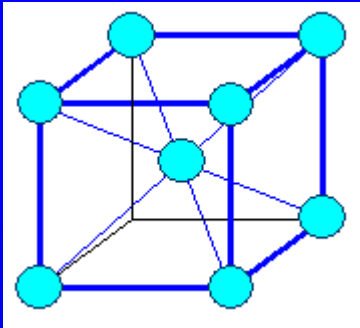
The forces among ions are electrostatic in origin.
 $\longrightarrow$ Ionic bonding is strong.

Since the filled shells of the ions are spherically symmetric, the ionic bonds in the solid are not directional.

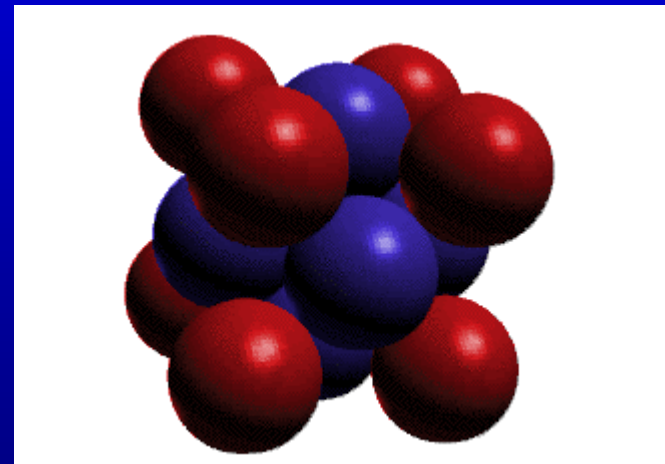
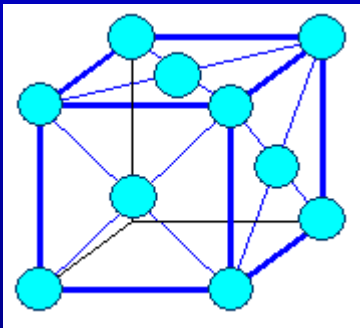
cubic lattice



simple cubic structure (sc)

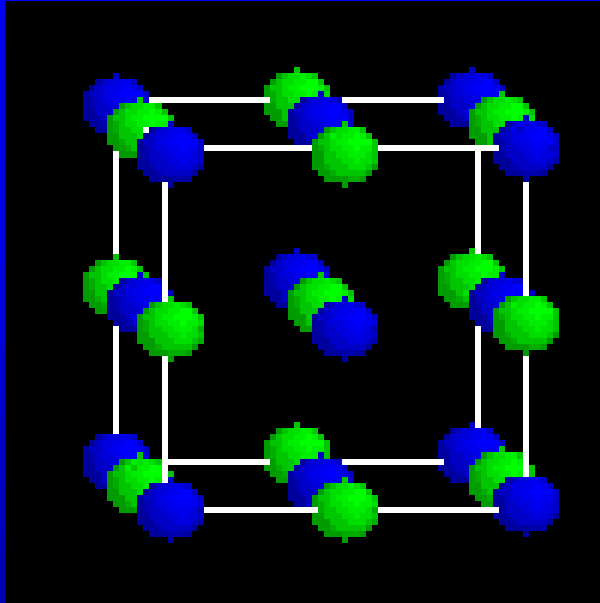


body-centered cubic structure (bcc)

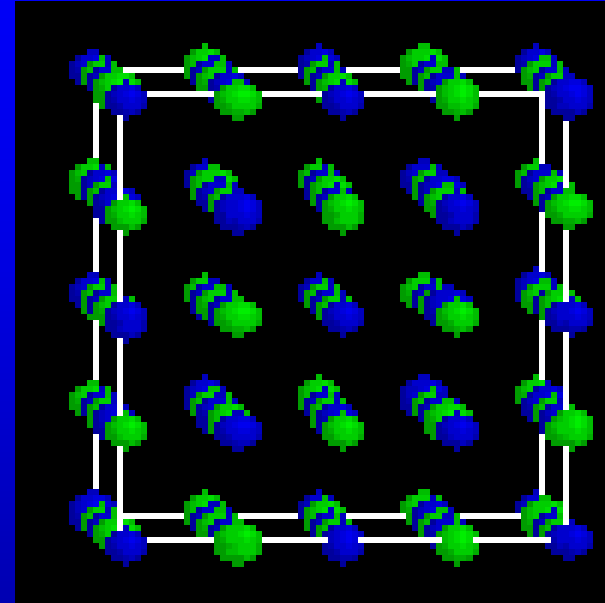


face-centered cubic structure (fcc)

The structure of the NaCl (sodium chloride) crystal



1 unit cell



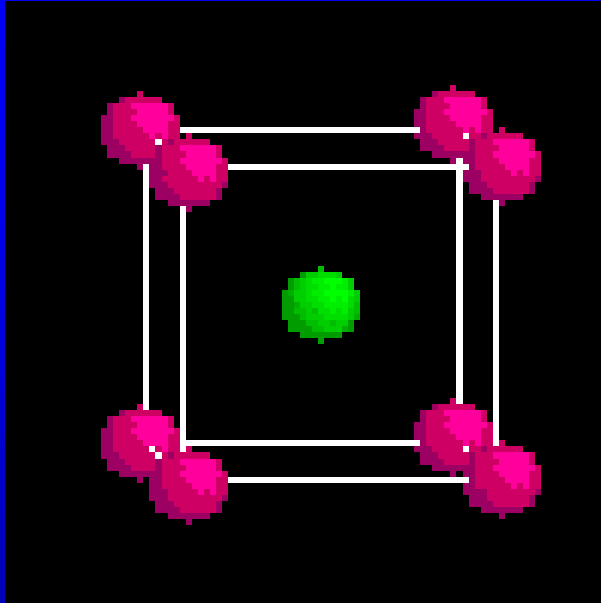
8 unit cells

Na^+ cations are the blue spheres.

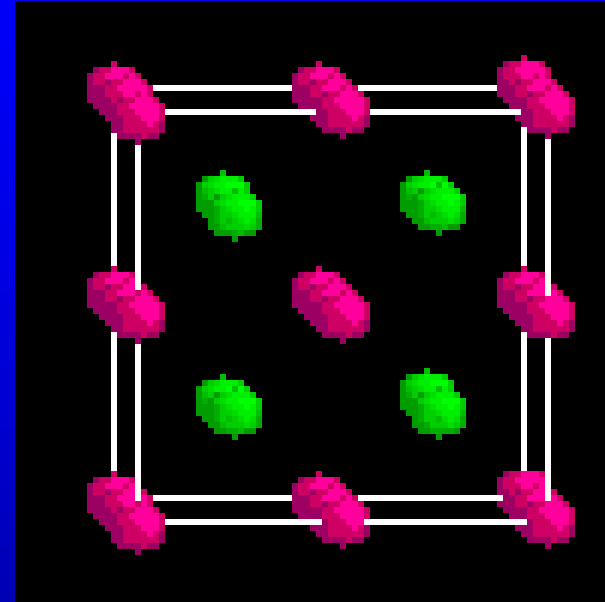
Cl^- anions are the green spheres.

Each Na^+ has six Cl^- for its nearest neighbors.

The structure of the CsCl (cesium chloride) crystal



1 unit cell



8 unit cells

Cs⁺ cations are the green spheres.

Cl⁻ anions are the pink spheres.

Each Cs⁺ has eight Cl⁻ for its nearest neighbors.

For some materials, the structure will change from fcc to bcc as the temperature is changed.

The potential is basically Coulombic

$$V_a = -\alpha \frac{1}{4\pi\epsilon_0} \frac{e^2}{r}$$

$\alpha = 1.747\ 6$ for fcc crystals;
 $\alpha = 1.762\ 7$ for bcc crystals

Pauli principle tends to keep those filled subshells from overlapping. This **means** that there exists a repulsive “potential”

$$V_r = \frac{A}{r^n}$$

crystal	n	structure
LiF	6	fcc
LiCl	7	fcc
NaCl	8	fcc
NaI	9.5	fcc
KCl	9	fcc
KBr	9.5	fcc
CsCl	10.5	bcc
CsI	12.0	bcc

$$V = -\alpha \frac{1}{4\pi\epsilon_0} \frac{e^2}{r} + \frac{A}{r^n}$$

$$\left. \frac{\partial V}{\partial r} \right|_{r_0} = +\alpha \frac{1}{4\pi\epsilon_0} \frac{e^2}{r_0^2} - n \frac{A}{r_0^{n+1}} = 0$$

$$\alpha \frac{1}{4\pi\epsilon_0} \frac{e^2}{r_0} \frac{1}{n} = \frac{A}{r_0^n}$$

$$-V_0 \equiv -V(r_0) = \alpha \frac{1}{4\pi\epsilon_0} \frac{e^2}{r_0} \left(1 - \frac{1}{n} \right)$$

ionic cohesive energy of the solid

ionic cohesive energy of the solid-----the energy needed to take the solid apart into positive and negative ions.

atomic cohesive energy of the solid-----the energy needed to take the solid apart into neutral atoms.

Example Calculate the atomic cohesive energy of the NaCl crystal.

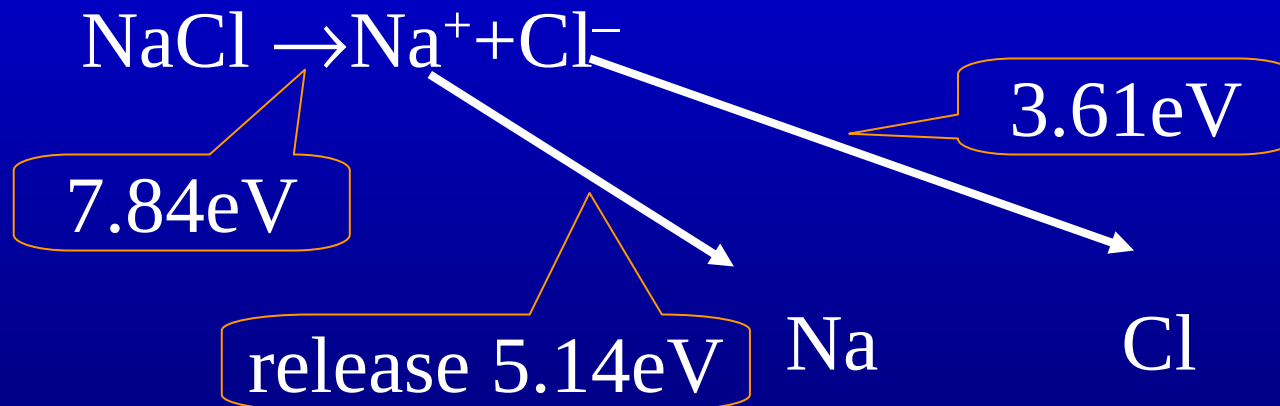
Solution: From the table, we know that $r_0 = 0.281 \text{ nm}$ and $n=8$ for NaCl. From the Eq.(28.3.4), The ionic cohesive energy of NaCl is

$$V_0 = -(1.7476) \frac{1}{4\pi\epsilon_0} \frac{e^2}{0.218 \text{ nm}} \left(1 - \frac{1}{8}\right) \\ = -7.84 \text{ eV}.$$

This represents the energy needed to remove Na^+ and Cl^- ions from the solid. The ionization energy of Na atom is 5.14 eV and the electron affinity of Cl is 3.61 eV . Then, the energy needed to form the neutral atoms from the solid is

$$7.84\text{eV} - 5.14\text{eV} + 3.61\text{eV} = 6.31\text{eV}.$$

This is the calculated atomic cohesive energy of the NaCl solid. The measured value is 6.39eV.



- $f \sim 1/r^2$ more stable than molecular solids ($1/r^3$, $1/r^7$)
- poorer electrical conductivity than metallic solids
- (cohesive energy $\sim$ eV) vaporization temperature $\sim 40 \times 300\text{K}$,
actual values are more like 1,000 to 2,000K
- transparent to visible light (ions:filled shells)
- absorb infrared radiation ($\lambda = 32.6\mu\text{m}$)
- soluble in polar liquids

soluble solvable solution solvent

The force acted on an ion can be estimated as

$$F = -\frac{dV}{dr} = -\alpha \frac{1}{4\pi\epsilon_0} \frac{e^2}{r_0^2} \left(\frac{r_0}{r}\right)^2 \left(1 - \frac{r_0^{n-1}}{r^{n-1}}\right).$$

Suppose we displace the atom by a small distance Δr from r_0 .

$$r = r_0 - \Delta r, \quad \left(\frac{r_0}{r}\right)^q = \frac{1}{\left(1 - \frac{\Delta r}{r_0}\right)^q} = 1 + q \frac{\Delta r}{r_0}$$

$$F \approx -\alpha \frac{1}{4\pi\epsilon_0} \frac{e^2}{r_0^2} (n-1) \frac{\Delta r}{r_0} \equiv -k\Delta r$$

For NaCl, we estimate the force constant to be

$$k = \alpha \frac{n-1}{4\pi\epsilon_0} \frac{e^2}{r_0^3} = 1.75 \times \frac{7 \times (1.44 \text{ eV} \cdot \text{nm})}{(0.281 \text{ nm})^3} = 795 \text{ eV/nm}^2,$$

And the corresponding frequency is

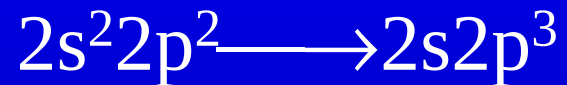
$$\nu = \frac{\omega}{2\pi} = \frac{1}{2\pi} \sqrt{\frac{k}{m}} = 9.20 \times 10^{12} \text{ Hz.}$$

(The mass of Na atom is used.)

$$\lambda = 32.6 \mu\text{m} \quad (\text{near infrared region})$$

Covalent solids

Covalent bonds are directional.



sp^3 hybridization

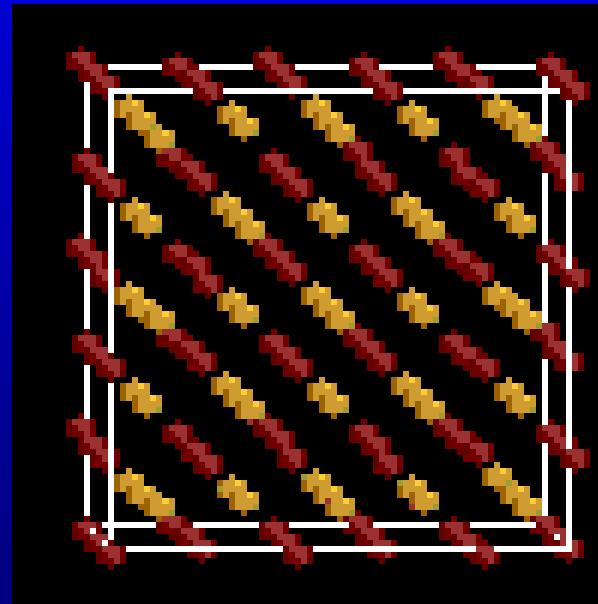
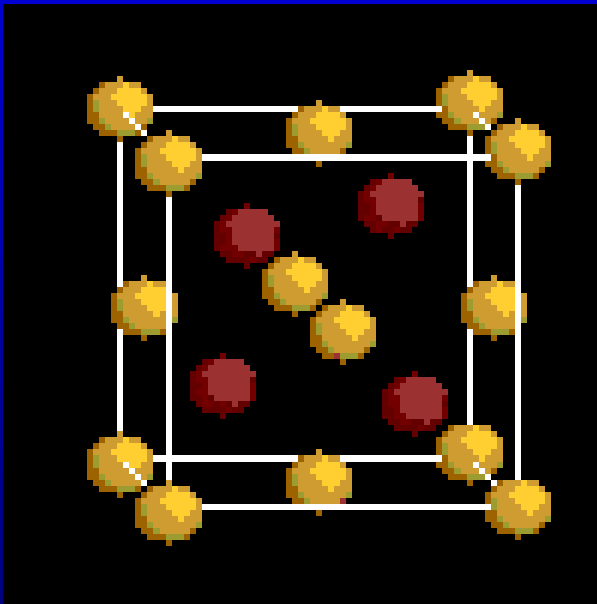
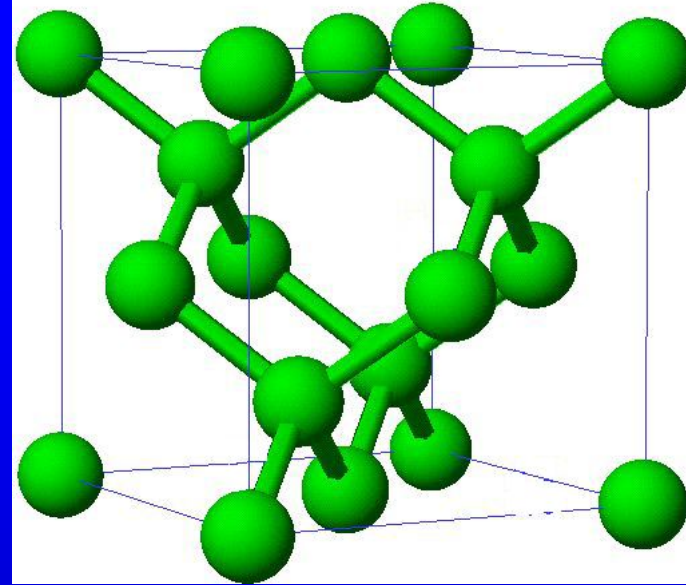
diamond

germanium (Ge) silicon (Si) ZnS (zinc sulfide)

SiC (silicon carbide, carborundum)



diamond



Zinc blende or sphalerite, ZnS

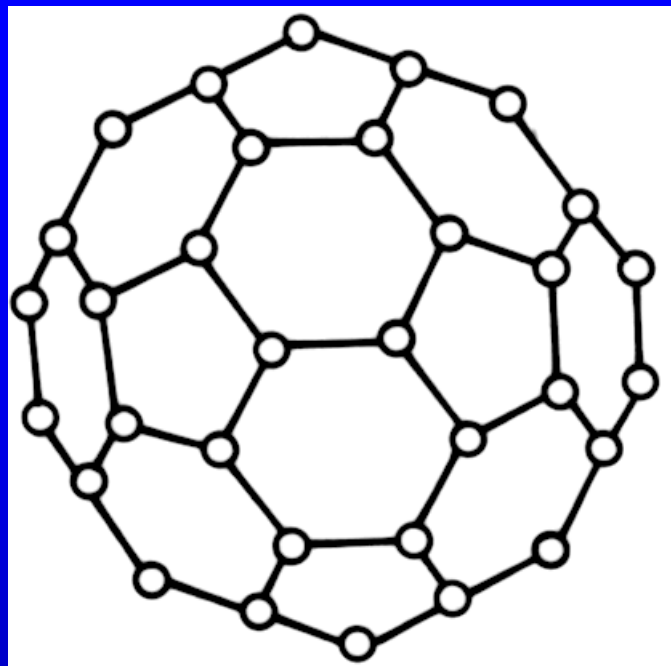
The covalent solid do not have the similarity of characteristics that ionic solids do.

Carbon, in diamond structure, has large bond energy and is then very hard and transparent to visible light.

Germanium and tin have the similar structures, but are metallic in appearance and highly reflective.

Melting point of diamond is 4000K.

But the melting points of Ge and Sn are lower than that of ordinary metals.



C_{60} looks like geodesic domes designed by American architect Buckminster Fuller.

Fullerene ball or buckyball

富勒球

巴基球

Buchyminsterfullerene (富勒烯)

1985, Harold W. Kroto(1930-), a British chemist and infrared and microwave spectroscopist

long chained carbon molecules in interstellar space

He theorized that these chains had been formed in the atmospheres of carbon-rich red giant stars.

Rice University

Robert F. Curl Jr. laser vaporization device

Richard E. Smalley mass spectroscopy

cage-like arrangement of carbon atoms

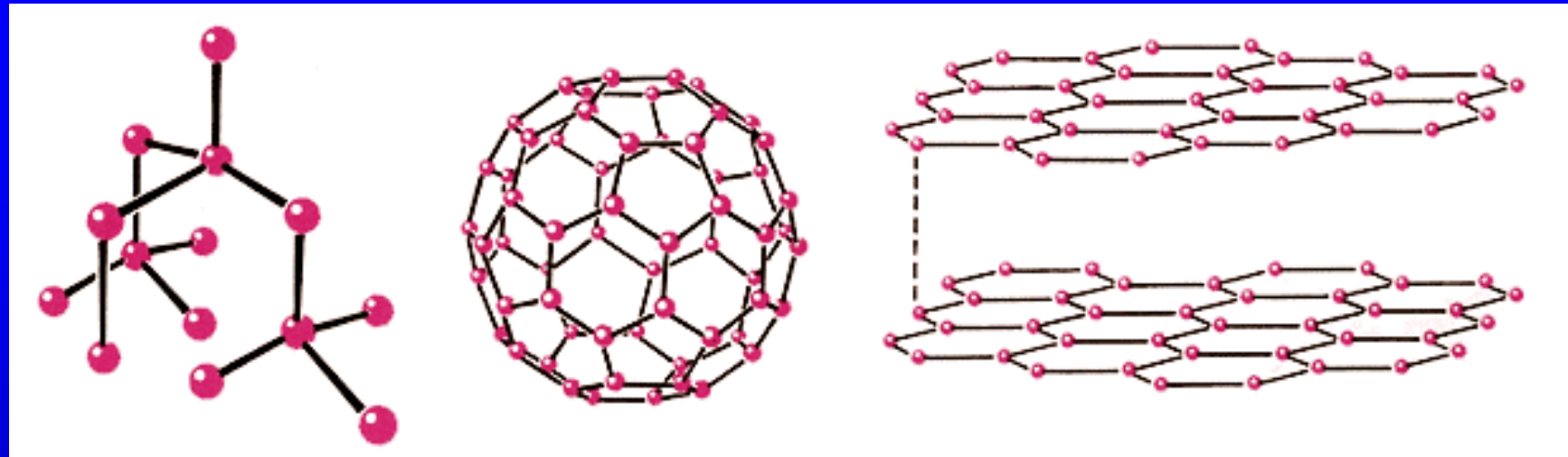
Making Fullerenes carbon arc

Helium

10 % of the soot should be made up of C₆₀

1 % C₇₀

transparent yellow solid



Graphite

diamond

amorphous
carbon(coke,
charcoal)

fullerenes

four allotropes of carbon



C70, C76, C78, C82, C84,...

C78--- C60 derived nanotube (buckytube)

Buckyonion

nanocone

Alkali-metal-doped crystals: M_xC_{60} (M = alkali metal, K, Rb) is superconducting $T_C \sim 28K$

Lucent scientists have inserted chloroform and bromoform between buckyballs and got them to superconduct at 117K.

Derivatives of buckyball have been found to be biologically active, and have been used to attack cancer.

Boris Yakobson et al. recently predicted the existence of boron buckyball-B₈₀.



Metallic bonds

Most common in bonding of metals and their alloys.

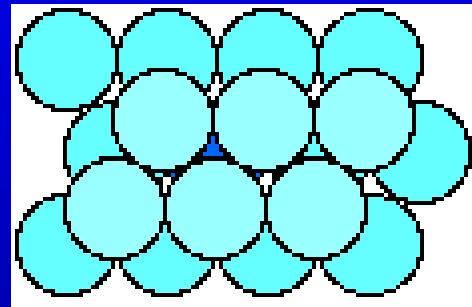
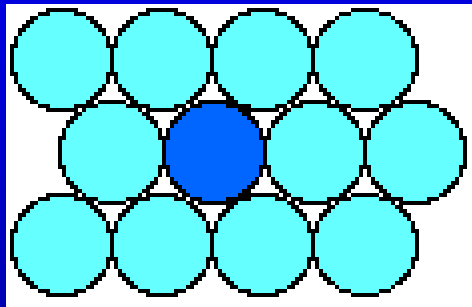
Proposed model of metallic bonding

metals usually have, at most, 3 valence electrons, all of which form an electron gas which drift through the entire metal.

Base electrons form net-positive ion cores, which attract the free electrons from the “sea” as needed to maintain neutrality.

The metal is held together by the attractive force between each individual metal ions and the electron gas.

The most common structures of metallic solids are fcc, bcc and hexagonal closed packed (hcp).



Molecular bonds

Molecular solids $\left\{ \begin{array}{l} \text{polar } f \sim 1/r^3 \\ \text{H}_2\text{O} \\ \text{non-polar } f \sim 1/r^7 \\ \text{inert gas, halogens,} \\ \text{and fullerball C}_{60}\dots \end{array} \right.$

polar dipole-dipole interaction

non-polar temporary fluctuating (instantaneous) dipoles

van der Waals interaction

Hydrogen bond

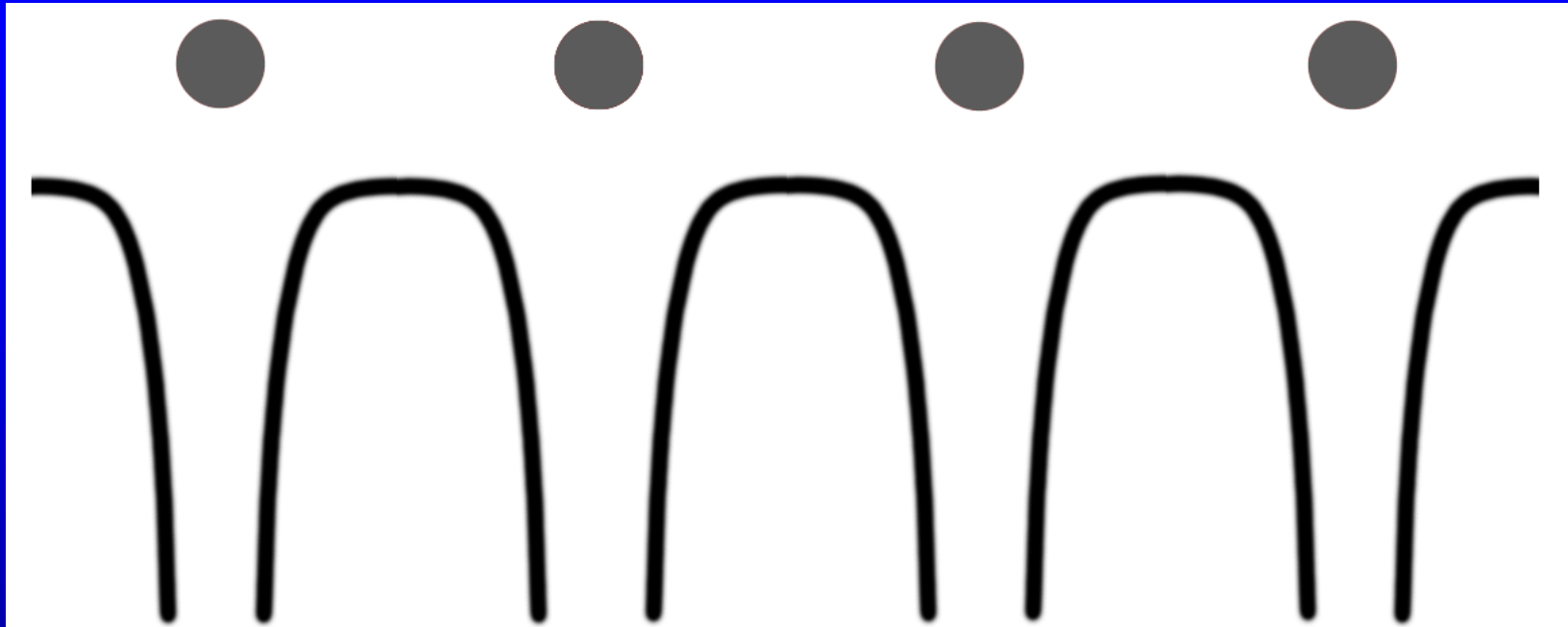
In a water molecule, the oxygen atom tends to attract all the electrons of the molecule, so looks like the negative end of the dipole, the two almost “bare” protons are the positive ends of the dipole, and each can attract the negative oxygen of adjacent water molecules.

hexagonal crystal structure of ice and then the hexagonal patterns of snowflakes



Actually, at room temperature water should be a gas and not a liquid, as are similar light molecules such as ammonia and hydrogen sulfide. (Yes, ammonia is a gas, the liquid which sometimes is called ammonia is actually a water solution of the gas.) The reason for water being a liquid and not as is the hydrogen bonds.

28.4 Free electron model for metals



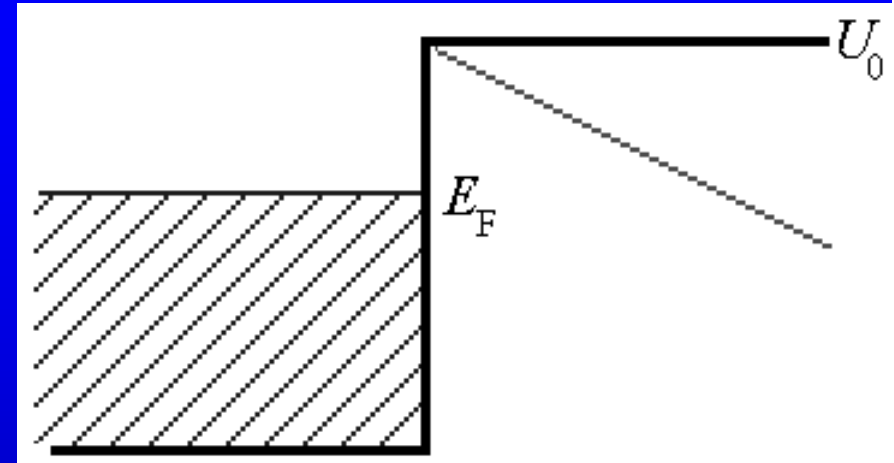
§ 27.4 Contribution of electrons to heat capacity of solids

§ 19.3 susceptibility with classical model

Thermal emission of electrons

$$W = U_0 - E_F$$

work function in § 22.2



The electrons can leave the well if

$$\frac{p_z^2}{2m} \geq W + E_F$$

$$E \geq \frac{p_x^2 + p_y^2}{2m} + W + E_F$$

quantum effusion(泻流)

$$d\Gamma = v_z \iint_{v_x, v_y} dn$$

$$= v_z \iint_{k_x, k_y} f(E) \frac{2 \cdot d^3 k}{(2\pi)^3}$$

$$= v_z \iint_{p_x, p_y} f(E) \frac{2 \cdot d^3 p}{(2\pi\hbar)^3}$$

$$\Gamma = 2 \int_{-\infty}^{\infty} \frac{dp_x}{2\pi\hbar} \int_{-\infty}^{\infty} \frac{dp_y}{2\pi\hbar} \int_{[2m(W+E_F)]^{1/2}}^{\infty} \frac{1}{e^{\beta(E-E_F)} + 1} \frac{\partial E}{\partial p_z} \frac{dp_z}{2\pi\hbar}$$

$$= \frac{2}{\beta h^3} \int_{W+E_F+\frac{p_x^2+p_y^2}{2m}}^{\infty} \frac{1}{e^{\beta(E-E_F)} + 1} dE \left[\rightarrow \frac{-\beta \left(W + \frac{p_x^2+p_y^2}{2m} \right)}{1+e^{-x}} \right] dp_x dp_y \frac{d(1+e^{-x})}{1+e^{-x}}$$

$$\beta W \sim 40/\text{eV} \times \text{eV} \sim 40 \quad e^{-(40+\dots)} \sim 10^{-18}$$

$$\Gamma \approx \frac{2}{\beta h^3} e^{-\beta W} \iint e^{-\frac{\beta}{2m}(p_x^2 + p_y^2)} dp_x dp_y$$

$$= \frac{2}{\beta h^3} e^{-\beta W} \left\{ \sqrt{\frac{2m}{\beta}} \int_{-\infty}^{\infty} e^{-x^2} dx \right\}^2$$

$$= \frac{4\pi m}{\beta^2 h^3} e^{-\beta W}$$

The emission flow is

$$J = -e\Gamma$$

In an external field
additional potential is $U = -eEx$,

the image potential

$$U = -\frac{e_s^2}{4x}$$

$$F = -\frac{dU}{dx} = -\frac{e_s^2}{4x^2} = -\frac{e_s^2}{(2x)^2}$$

Assignment: 28.7(optional)

28.5 Energy band

Bloch theorem:

for periodic potential, the solution of Schrödinger equation is of the following form

$$\psi_k(\mathbf{r}) = u_k(\mathbf{r}) \exp(i\mathbf{k} \cdot \mathbf{r})$$

where $u_k(r)$ is of the periodicity of the lattice.

For a model of 1D crystal of period a ,

$$\begin{aligned}\psi_k(x+a) &= u_k(x+a) \exp[ik(x+a)] \\ &= \exp(ika) u_k(x+a) \exp(ikx) \\ &= \exp(ika) u_k(x) \exp(ikx) \\ &= \exp(ika) \psi_k(x)\end{aligned}$$

Translational invariance requires

$$\psi(x + a) = \lambda(a)\psi(x), \quad |\lambda(a)| = 1$$

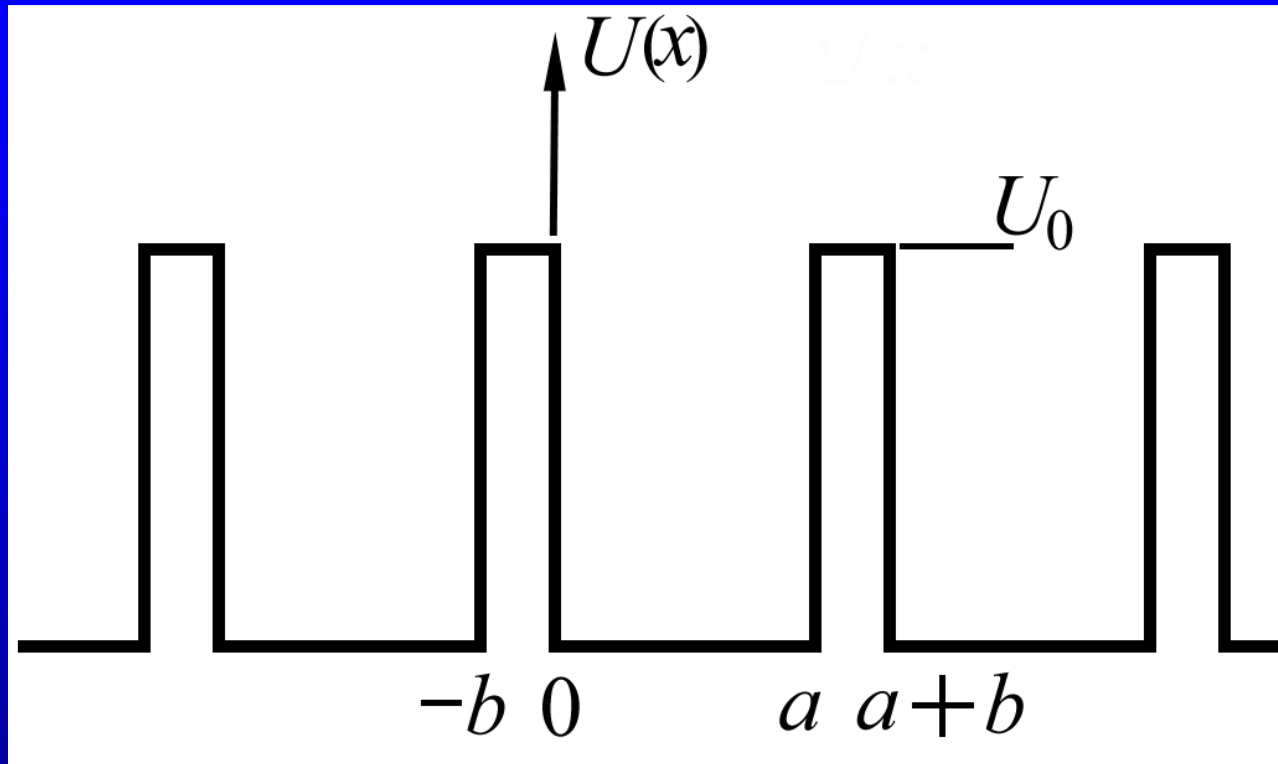
Two translations lead to

$$\begin{aligned}\psi(x + a' + a'') &= \lambda(a' + a'')\psi(x) \\ &= \lambda(a'')\psi(x + a') \\ &= \lambda(a'')\lambda(a')\psi(x)\end{aligned}$$

So

$$\lambda(a) = \exp(ika)$$

Kronig-Penny model



$$\psi(a < x < a + b) = \psi(-b < x < 0) \exp[ik(a + b)]$$

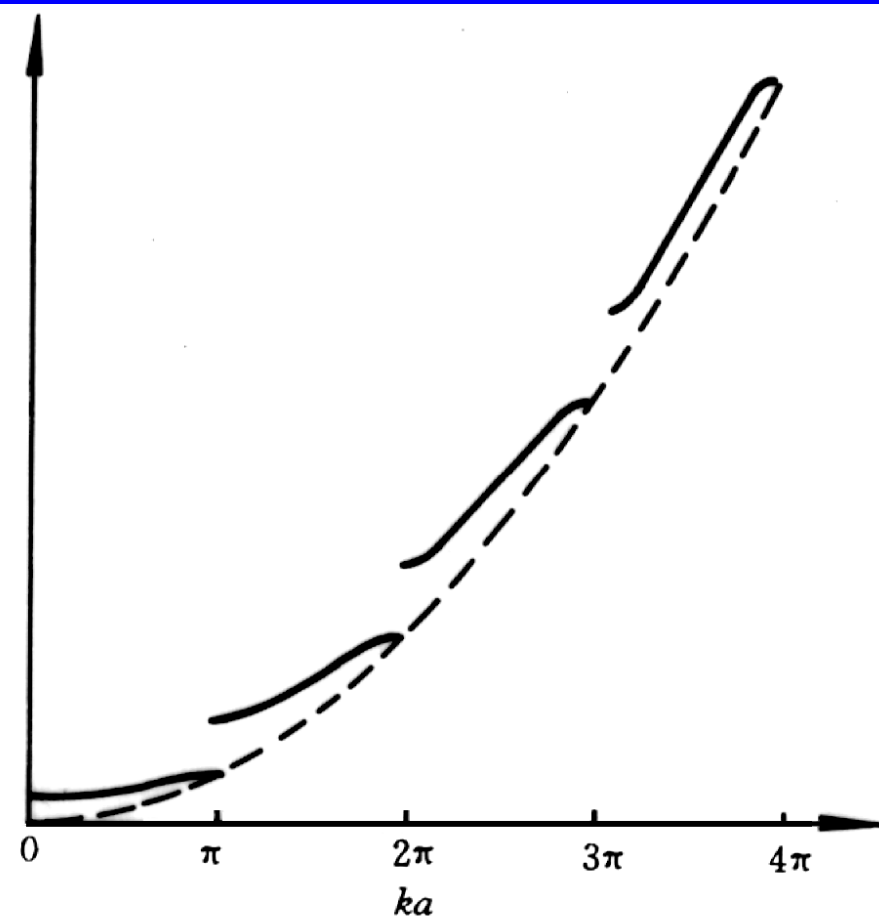
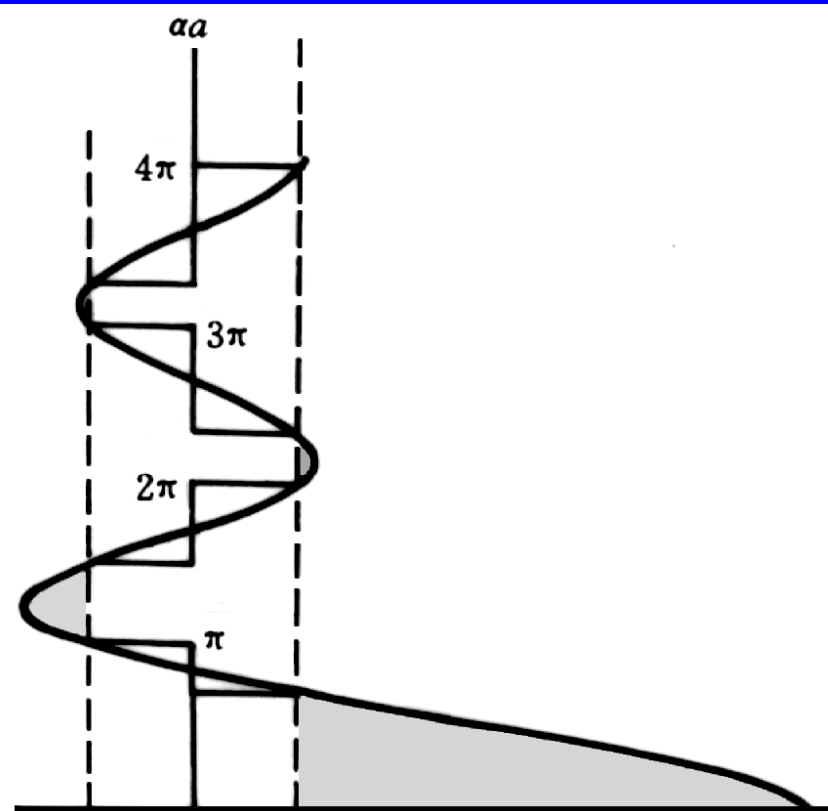
Standard quantum mechanics procedure will lead to k - E relation:

$$\cos ka = \frac{mabU_0}{\hbar^2} \frac{\sin \alpha a}{\alpha a} + \cos \alpha a$$

$$\alpha = \sqrt{\frac{2mE}{\hbar^2}}$$

$$U_0 = 0, k = \alpha. \quad E = \hbar^2 k^2 / 2m$$

方俊鑫，陆栋， 固体物理学（上册），上海科学技术出版社（1980）。



rhs of Eq(28.5.3)